

Counterparts

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Draft of 9th March 2015
Comments welcome

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Chapter 1

A-theories and B-theories

1.1 Two key questions about time and modality

The central faultline in the philosophy of time is between the ‘A-theory of time’ and the ‘B-theory of time’. (This singularly unhelpful terminology is the fault of McTaggart 1908.) Since this book is devoted to working out a view whose classification either as an A-theory or as a B-theory is difficult, we will need to be careful in characterising what is at stake. But let us begin with some slogans and picture thinking, to help convey an intuitive sense for the two positions.¹

<i>The A-theory of time</i>	<i>The B-theory of time</i>
The present time is metaphysically special.	All times are “on a par”.
There are facts about which times are past, present and future, over and above the facts about which times are before which.	There are no such further facts. Pastness, presentness and futurity “reduce” to beforeness and afterness.
Time genuinely “flows” and “passes”. There is “objective becoming”.	We live in a “block universe”.
The facts themselves change.	Change is just a certain kind of pattern in the facts.

There is a lot of murkiness here. Rather than spending a long time parsing the slogans, I will proceed directly to a pair of questions in the vicinity which I think are clear enough to make progress with. Having got the questions on the table and made a case that they are intelligible and worth debating,

¹(Obviously I will need lots of citations here. . . .)

we will return in section 1.6 to the questions whether there is any important further doctrine in the vicinity of the A-theory that does not follow from answering both questions positively, and whether there is any important further doctrine in the vicinity of the B-theory that does not follow from answering both questions negatively.

The first question is

T1 Are there temporarily true propositions?

Call those who answer ‘yes’ to T1 ‘propositional temporalists’, and those who answer ‘no’, ‘propositional eternalists’.

Propositional temporalism is intended to gloss the last of our A-theoretic slogans, ‘The facts themselves change’. The inference from the slogan to propositional temporalism seems straightforward. Surely we can agree that the proposition that *P* is true when, and only when, the fact that *P* obtains; so if the fact that *P* is one of the facts that change—if, that is, the fact that *P* obtains only some of the time—then the proposition that *P* must be true only some of the time. The reverse inference, from propositional temporalism to the slogan, is obviously unproblematic if we take ‘fact’ to be just another word for ‘true proposition’. But to sustain the inference, we only need the much weaker thought that the true propositions which *do* correspond to facts jointly entail all the other true propositions. For in that case, any time the truth value of a proposition changed, there would have to be some change in the totality of obtaining facts.² And this thought should be acceptable even to proponents of “sparse” theories of facts, according to which the only facts are facts about the instantiation of certain “fundamental” properties and relations. For it seems to be part of the point of the idea of fundamental properties and relations that all truths are entailed by the totality of truths about which things there are and which fundamental properties and relations they instantiate. (Section 1.5 below will revisit the question whether there is a coherent interpretation of the slogan that does not follow from propositional temporalism.)

²I am assuming here that if $p_1 \dots p_n$ entail p_{n+1} —i.e. that if it is metaphysically necessary that p_{n+1} is true if $p_1 \dots p_n$ are—then it is *always* the case that p_{n+1} is true if $p_1 \dots p_n$ are true. This seems obvious: what is metaphysically necessarily the case is always the case. But it is, surprisingly, controversial. In some combined logics for tense and modality, for example Kaplan’s “logic of demonstratives” (Kaplan 1989), ‘necessarily *P*’ does not entail ‘always *P*’. The considerations which might lead someone to deny this inference are discussed at length in Dorr and Goodman n.d.

T1 is not a question about language. But it is intimately associated with a debate in the philosophy of language, about the kind of semantic account appropriate for a certain large class of ordinary sentences. The sentences in question are the *temporally variable* sentences: those which could be used literally at two different times in such a way that one utterance involved someone saying something true, and the other involved saying something false—‘Obama is president’, for example. Given propositional eternalism, the semantic functioning of these sentences cannot be helpfully captured by choosing a *single* proposition to call their “semantic value” (in the relevant language). The reasons for this are well expressed by Frege, the archetypal propositional eternalist:

But are there not thoughts which are true today but false in six months time? The thought, for example, that the tree there is covered with green leaves, will surely be false in six months time. No, for it is not the same thought at all. The words ‘This tree is covered with green leaves’ are not sufficient by themselves for the utterance, the time of utterance is involved as well. Without the time-indication this gives we have no complete thought, i.e. no thought at all. Only a sentence supplemented by a time-indication and complete in every respect expresses a thought. But this, if it is true, is true not only today or tomorrow but timelessly. (Frege 1918, p.)

To give a non-crazy semantic account of sentences like ‘The tree is covered with green leaves’ and ‘Obama is president’, propositional eternalists must classify them as *context-sensitive*. Like ‘I am a philosopher’ and ‘This is a cup’, they are flexible tools whose meaning allows them to be used to express many different propositions. Some literal utterances of ‘Obama is president’ express eternally true propositions, while others express eternally false propositions; no single proposition is expressed by every utterance of the sentence. I will call the claim that all temporally variable sentences are context-sensitive in this sense “temporal contextualism”.

Note that temporal contextualists are free to give a range of different answers to the question which propositions are expressed by particular utterances of ‘Obama is president’. The most obvious candidates are propositions, concerning some particular instant *t* or interval *T*, to the effect that Obama is president at *t* or throughout *T*. However, this ‘date theory’ is certainly not the only option for the temporal contextualist. According to another time-honoured view, the ‘token-reflexive theory’, a particular utter-

ance u of 'Obama is president' will express the proposition that Obama is president during u .

Propositional eternalists, as we have seen, are pretty much required to be temporal contextualists. Propositional temporalists will not be persuaded by the argument given above: it would be natural for a propositional temporalist to hold that 'Obama is president' context-insensitively expresses a single proposition which is true exactly when Obama is president. Of course, propositional temporalists might have some other reason for positing context-sensitivity in this sentence; they might even hold the radical view that *all sentences whatsoever* are context-sensitive. But their belief in temporarily true propositions at least opens up the possibility of a very different account of the range of propositions available to be expressed by such sentences. This makes a broad range of linguistic considerations potentially relevant to T1.

Nevertheless, it would be a mistake to think of T1 as merely a question about the best way to do semantics. Unlike 'semantic value', 'proposition' is not just a piece of jargon which different semantic theories are free to use as they see fit. It is firmly anchored in ordinary discourse by certain rules which license us in treating a wide range of sentences involving 'proposition' as equivalent to certain other sentences not involving 'proposition': for example, 'The proposition that snow is white is true' is to be treated as equivalent to 'Snow is white'. It is these anchoring rules that make T1 a question of general interest to metaphysics, as opposed to a question *merely* about the particular ontological category of propositions. Section 1.2 will attempt to systematise the rules in question, and to explain how they can be acceptable to both propositional temporalists and propositional eternalists.

The second question by which I propose to capture the A-theory/B-theory dispute is T2:

T2 Is the unit class of the present instant much *more natural* than the unit classes of most other instants?

Call those who answer 'yes' to T2 *temporal elitists*, and those who answer 'no' *temporal egalitarians*. Temporal elitism is intended to capture A-theoretic slogans like 'The present is metaphysically special', while temporal egalitarianism captures B-theoretic slogans like 'All times are on a par'.

The expression 'natural class' in T2 is used in the sense of Lewis 1983a: it contrasts with 'gerrymandered class'. The idea is best introduced by means of examples. The class of all electrons is very natural. The class of all

electrons except for the ones in my left toe together with all donkeys is much less natural. A natural class groups things together in a non-arbitrary way: it carves nature at its joints. According to the temporal elitist, a classification of the instants of time into (i) the present instant; (ii) past instants (those that come before the present); and (iii) future instant (those that come after the present) carves along a natural joint. It is much more natural than the following classification: (a) the instant at which my alarm clock started going off yesterday morning; (b) instants earlier than that instant; (c) instants later than that instant.³

In using the language of naturalness, I am not presupposing any particular metaphysical account of what the naturalness of a class consists in. Nor am I presupposing anything about the importance of such facts to the project of metaphysics. *Prima facie*, claims about relative naturalness seem to be somewhat vague; it is not obvious that such vague language is especially useful for the project of developing a systematic metaphysical theory. Its main use, rather, is similar to that of the expression ‘metaphysically special’ for which it serves as a gloss: it is supposed to facilitate comparison of different metaphysical theories, by providing a theoretically neutral label for a structural feature that could be shared by such theories even when they are otherwise very different, and employ very different theoretical primitives. Expressions like ‘privileged’ and ‘distinguished’ often play a similar role, as when people characterise relativity theory as including the idea that no frame of reference is “privileged”. The idea is to reject a large family of theories which can be seen as agreeing that there is a privileged frame of reference, while saying different things about what its privilege consists in (e.g. being picked out by the points of absolute space, being the frame in which ether-particles are at rest, being a frame for which frame-relative simultaneity coincides with absolute simultaneity . . .). In view of the pervasive role of such expressions in physics, the idea that all claims of comparative naturalness are false or unintelligible is an extremely radical one.

Why have I set up T2 as a comparison between the present instant and *most* other instants? The point was to not to rule out the possibility that there are some other “distinguished” instants in the time-series which can be singled out in a non-arbitrary way. For example, if there is a first instant or a last instant, they are obviously quite special—their unit classes are quite natural—and likewise, if there is a unique instant that is exactly halfway

³Note that the notion of naturalness I am working with is one on which the *complement* of a very natural class is itself quite natural, since it carves along the same joint.

between the first instant and the last instant, its unit class is pretty natural too. Other instants may be special in virtue of what happens at them—e.g. the end of the era of cosmic inflation, or the beginning of the Rapture, or what have you. While many A-theorists will probably want to say that *being present* makes for *more* naturalness than any of the other features that uniquely distinguish particular instant, this strong claim is hard to argue for, and its denial is clearly far too weak to vindicate the B-theoretic slogan that “all times are on a par”. For our purposes the vague ‘most’ claim looks to be of about the right strength to divide those who think that the present is “privileged” from those who think that it is “just one time among many”.

It is tempting to get around this issue by introducing a more expressive language for talking about naturalness. We might, for example, allow ourselves to speak of classes as being natural in different *respects*; we could then characterise the A-theory/B-theory debate as turning (at least in part) on the question there there is a *respect* in which the unit class of the present instant is *uniquely* natural. But the new ideology of “respects” raises many questions that it would be nice to avoid; and in any case, the negative answer to this question does not do a good job of capturing the B-theoretic slogan that “all times are on a par”, since it can be accepted by an annoying character who accepts that there is a respect in which the unit class of the present instant is uniquely natural but only on the boring grounds that *every unit class whatsoever* is uniquely natural in its own distinctive respect.⁴

Another worry about T2 as a device for capturing the A-theory/B-theory issue is that some B-theorists will answer ‘yes’ if you happen to catch them at the right time. For example, if you email a B-theorist asking T2, a B-theorist who thinks that the Rapture will occur at exactly 8.32 pm this evening might type ‘yes’ in response and strive to press ‘send’ at exactly 8.32 pm, reasoning that this will be a way of communicating the truth that the unit class of

⁴Another strategy for getting around problems associated with “distinguished” instants other than the present is to talk about natural *properties* rather than natural *classes*. We could characterise A-theorists as thinking that *presentness* is a very natural property, and B-theorists as denying this. The problem with this is that there is disagreement within the “all times are on a par” camp about how to talk about *properties*. Some (***cite adverbialists***) think of properties as things that need to be supplemented with an object *and a time* before we have a complete proposition. Those who talk like this might well agree that the property of presentness is extremely natural: it has the structurally very distinctive feature that necessarily, it yields a truth whenever you supplement it with the same instant of time twice over, and a falsehood otherwise. Focusing on the naturalness of classes avoids this distracting issue, since no-one thinks that the question whether an object belongs to a class “only makes sense relative to a time”.

8.32 pm is much more natural than the unit classes of most other times. To avoid this worry, we might be tempted to modify T2 by adding a temporal modifier, as follows:

(T2*) Is it *always* the case that the unit class of whichever instant is present is much more natural than the unit classes of most other instants?

The disadvantage of this formulation is that some people who clearly belong on the B-theoretic side of our classification will be tempted to answer 'yes' for the wrong reasons. When a sentence occurs in the scope of a tense operator like 'always', there is strong interpretative pressure to find a way of understanding the sentence in such a way that the operator is not redundant. This pressure might prompt a B-theorist to reinterpret 'natural' as it occurs in (T2*) as the kind of predicate that can interact non-trivially with 'always'—in one common framework, that would mean positing a hidden time variable bound by 'always' as one of the arguments of the predicate. Various interpretations of 'natural relative to t ' will suggest themselves to the creative. Where x is some entity (perhaps utterly undistinguished), we could say that class A is natural relative to x iff there is some natural class of ordered pairs O such that A comprises all and only those objects y for which $\langle x, y \rangle$ belongs to O . For example, if x is a blob of mud with a specific hard-to-describe blobby shape, the class of all objects with exactly that shape will not be very natural *simpliciter*, but it will be very natural *relative to x* , since the class of ordered pairs whose two elements are exactly the same shape is very natural. Since the class of all ordered pairs that consist of the same object twice over is (intuitively) very natural, this definition of relative naturalness strongly suggests that relative to any object x , the unit class of x is very natural. Thus, assuming they agree that every instant is present at itself (and why wouldn't they?), even B-theorists will give a positive answer to (T2*) if they interpret it as 'Is every instant t such that the unit class of the instant that is present at t is more natural *relative to t* than the unit classes of most other instants?' For this reason I think we will do better to focus on T2, which does not cry out in the same way for an interpretation in terms of a notion of naturalness relative to t . For our present purposes, it is more important to minimise the opportunities for misunderstanding than to come up with a formulation of the question that works even in the presence of bizarre hypotheses according to which some of the times at which we are having this discussion play some *permanently* distinguished role within the time series.

Questions T1 and T2 only succeed as articulations of basic questions about the metaphysics of time if we hold fixed certain background ontological assumptions, about *propositions* in the case of T1, and *instants of time* in the case of T2. When people reject these assumptions, their answers to T1 and T2 may reflect only their heterodox views about propositions and instants, rather than anything that is intuitively relevant to their categorisation as A-theorists or B-theorists. For example, nominalists, who deny the existence of propositions and/or instants of time on the grounds that there are no abstract entities of any sort, could still be recognisable either as A-theorists or as B-theorists, even though they will be unable to use T1 and T2 to articulate these views (unless they have some story about why it is sometimes legitimate to play along with ordinary ways of talking about abstract entities). This sort of situation is very common in philosophy. It is tempting to react to it by looking for perfectly neutral formulations of the relevant questions, whose answers will depend only on the subject matter we are concerned with without the need for any background assumptions. But since such neutrality is typically unachievable, the best way to pursue clarity is not to dispense with the need for any background assumptions, but to articulate the required assumptions as explicitly as possible. Section 1.2 below will attempt to do this for the background theory of propositions presupposed by T1, while section 1.3 will attempt to do the same for the background theory of instants presupposed by T2. Once the required assumptions are on the table, we will at least be able to tell whether a given theorist rejects the assumptions, and will hopefully be better equipped to tackle the delicate enterprise of figuring out how to rephrase our questions in a such a way as to capture the underlying metaphysical issues in a form accessible to the theorist in question. If we are lucky, we will be able to find some systematic replacements for the distinctive vocabulary that occurs in our background theory of propositions ('proposition' and 'true'), or instants ('instant', 'past', 'present', 'future', 'before') such that the background theories come out true once the replacements are made, according to the theorist. In such cases we will likely find that the best way to classify our theorist is simply to look at their answers to the questions that result from making the relevant replacements in T1 and T2.⁵

⁵It is worth noting that temporal elitists and egalitarians are likely to disagree about the naturalness of many classes other than classes of instants. For example, in a world where objects often change their shape, it would be plausible for a temporal elitist to hold that the class of all *currently* spherical objects is much more natural than most other classes that have

Since we have two yes-no questions, there are in principle four combinations of views we need to think about. Two of the combinations—propositional temporalism plus temporal elitism, and propositional eternalism plus temporal egalitarianism—have a familiar feel; they are part of the stereotypical A-theoretic and B-theoretic packages. But must we also take seriously the hybrid views which answer ‘yes’ to one question and ‘no’ to the other? Although none of the central claims in the book will depend on this, I think there are strong arguments that these hybrids are non-viable.

The case against the combination of propositional eternalism and temporal elitism is fairly straightforward. Clearly, the temporal elitist will think that one can speak truly by uttering ‘The unit class of the present instant is more natural than those of most instants’ no matter when it is. So, for example, utterance ‘{2015} is more natural than {2014}’ will express truths in 2015 and falsehoods in 2014. As we have seen, propositional eternalists have to account for this kind of temporal variability in language by positing context-sensitivity. In the present case, the relevant source of context-sensitivity would be ‘more natural than’. But treating ‘more natural than’ as context-sensitive in *this* way would completely defeat the intended purpose of the predicate. It’s just as if a believer in Minkowski spacetime were to understand ‘privileged frame of reference’ as context-sensitive in such a way that anyone who says ‘There is exactly one privileged frame of reference, and right now I am at rest relative to it’ thereby speaks the truth. (The same point applies to ‘metaphysically special’, ‘distinguished’, etc.) It is central to the communicative role of these expressions that we avoid interpreting them as having this kind of extreme context-sensitivity; there is no point in using the expressions at all if our interlocutors are unwilling to exercise this minimal level of charity.

The combination of propositional temporalists and temporal egalitarianism has been much more popular, and my argument against it will be more controversial. It goes as follows. Say that an instant is ‘accurate’ iff the propositions that are true *at it* are all and only the *true* propositions. If propositional eternalism is true, all instants are accurate, since a proposition that is true at one instant is true at every instant. By contrast, if propositional temporalism is true, most instants are inaccurate. The sun is either shining or

in the past, or will in the future, contain all and only the spherical objects. By focusing on disagreements like this one, we could avoid the need for any assumptions about the ontology of instants. But it is not clear how, without mentioning instants, we could abstract away from the details of this example to a general thesis.

not. If it is, the proposition that the sun is shining is true, and all the instants at which it is false are inaccurate; if it is not, the proposition that the sun is not shining is true, and all the instants at which *it* is false are inaccurate. Among the instants that are accurate as regards whether the sun is shining, many are inaccurate as regards whether I am sitting. Clearly, at most a few instants are accurate in every respect: in fact, there is a strong case that the present instant is the *unique* accurate instant. On its face, accuracy seems like a very interesting feature for an instant to have. A classification that puts the one and only accurate instant in a category by itself seems to do a much better job of carving at the joints than one that, say, puts the instant that will be accurate 17 years from now into a category by itself. Thus accuracy would seem to be a rather natural property, and the class of all accurate instants would seem to be a rather natural class.

Given the popularity of the combination of propositional temporalism with temporal egalitarianism, this line of argument deserves some scrutiny. Section 1.4 will develop it more carefully, while section 1.5 will consider ways in which it might be resisted.

Our questions about the metaphysics of time have modal analogues:

M1 Are there contingently true propositions?

M2 Is the unit class of the actual world much more natural than the unit classes of most possible worlds?

Call those who answer 'yes' to M1 *propositional contingentists*, those who answer 'no' *propositional necessitarians*; those who answer 'yes' to M2 *modal elitists*, and those who answer 'no' to M2 *modal egalitarians*. I will also sometimes call those who answer 'yes' to both questions "modal A-theorists", and those who answer 'no' to both, "modal B-theorists".

One kind of propositional necessitarian is the Spinozist, who denies that we can ever speak truly by saying something of the form 'It is contingent whether ϕ ', or 'Not necessarily ϕ , and not necessarily not- ϕ '. But I will mostly be thinking about a different sort of propositional necessitarian, whose view is more closely modelled on the standard B-theoretic package in the philosophy of time. Just as standard propositional eternalists hold that 'It is only sometimes the case that ϕ ' can be true without there needing to be any proposition that is only sometimes true, these propositional necessitarians hold that 'It is contingent whether ϕ ' can be true without there needing

to be any proposition that is contingently true. Not being Spinozists, they agree that there are many possible worlds. Just as propositional eternalists must posit context-sensitivity in a sentence ϕ whenever there are two different times at which ϕ is uttered literally, at only one of which the person uttering ϕ thereby asserts a truth, so propositional necessitarians must posit context-sensitivity in ϕ whenever there are two different *possible worlds* at both which ϕ is uttered literally, such that only one of them is a world at which the person uttering ϕ thereby asserts a truth. For example, a propositional necessitarian will take the sentence 'Jupiter and Saturn never collide' to be context-sensitive: it is capable of being used, literally, to express both (necessary) truths and (necessary) falsehoods. The most straightforward account of this context-sensitivity would be as follows: at any given possible world w , people who utter 'Jupiter and Saturn never collide' thereby express the proposition that Jupiter and Saturn never collide *at w* : a proposition that, according to the standard way of talking about possible worlds, is necessary if true, and impossible if false.

(Some philosophers that propositions are "individuated by their truth conditions", i.e. that whenever it is necessary that proposition p is true iff proposition q is true, p is identical to q . The combination of this doctrine with propositional necessitarianism entails that there are only two propositions: one necessarily true, the other necessarily not true. This is quite bizarre, and hard to reconcile with the idea that propositions are "objects of the attitudes", an idea which will play an important part in the proposition role as we will lay it out in section 1.2. Thus, propositional necessitarians will probably want to adopt some more "fine-grained" account of the sufficient conditions for identity among propositions. This could, for example, be a Russellian conception of propositions as complexes with a structure parallel in some respects to that of the sentences that express them, or a Fregean conception on which propositions somehow build in the criteria people need to satisfy in order to bear propositional attitudes towards them. At any rate, it would be a mistake to assume that propositional necessitarianism has to go along with the view that there are only two propositions.)

The arguments sketched above that T1 and T2 should be answered in the same way have parallels for M1 and M2. From propositional contingentism to modal elitism: given propositional contingentism, it seems like we should be able to pick out the actual world as the unique *accurate* world; and this distinction seems quite an interesting one, metaphysically speaking; just the sort of thing that should make for a "joint in nature" between the actual

world and all the inaccurate world. From modal elitism to propositional contingentism: if we consider only its non-contingent characteristics, the actual world is not all that special; any facts about it that could single it out as special would have to be contingent ones. As in the temporal case, the latter inference has to deal with interference arising from other kinds of natural distinctions that could, in principle, be imputed to the actual world even by a propositional necessitarians. For example, the propositional necessitarian might hold, at a certain world w , that w necessarily enjoys the supreme distinction of being the best of all possible worlds. At every world, the wise realise that w is best, and that its unit class is quite natural. At w , uses of 'the actual world' refer to w , and those who are wise at w know this at w . So at w , the wise answer 'yes' to M2, despite being propositional necessitarians. This corresponds to the worry that certain propositional eternalists might answer 'yes' to T2 if you happen to ask them at an instant they regard as very special, e.g. at the instant at which they take the Rapture to begin. However, the issue is harder to dismiss in the modal case, since we can't get around the problem just by asking M2 again. Still, I think we will do better to tolerate this potential to misclassify those with eccentric views rather than attempting to elaborate the wording of M2 to control for the issue: the elaborations are apt to do more harm than good, by presenting new interpretative pitfalls even for those with non-eccentric views.

Despite the formal analogies between T1 and M1, and between T2 and M2, there is (at least) a salient sociological difference. In the temporal case, the B-theory is the dominant view, but A-theorists are a vocal minority. In the modal case, by contrast, almost everyone who has considered the question has come down on the side of propositional contingentism and modal elitism. (And there is no point in my making a secret of the fact that I am entirely in agreement with this near-consensus.) There is thus much to be learnt by considering the two debates in parallel, calibrating our assessment of the weight of any given consideration by comparison with its analogue in the other debate.

The best-known dissenter from the consensus in favour of modal elitism is Lewis (1986b). Lewis famously denies that there is any important metaphysical distinction between the thing he calls 'the actual world' and the other things he calls 'possible worlds'. However, it would be a mistake to take Lewis as our paradigm of a "modal B-theorist". For one thing, he endorses propositional contingentism. For another, as we will see in section 1.3, the role played in Lewis's theory by the things he calls 'possible

worlds' does not perfectly fit the standard "possible world role". So we must be on the alert lest we mistake some verbal difference regarding the use of this particular piece of philosophical jargon for the substantive dispute we are trying to zero in on.

The rest of this chapter will be organised as follows. First, section 1.2 will explain the background theory of propositions assumed by T1 and M1, and explain why it is not enough on its own to settle these questions. ?? will digress into the philosophy of language, carefully laying out the argument sketched above that propositional eternalists and necessitarians are committed to regarding certain kinds of sentences as context-sensitive. Section 1.3 will then turn to T2 and M2, laying out the background assumptions about instants of time and possible worlds which these questions assume. Section 1.4 will revisit the argument given above for thinking that these questions should be answered in the same way as T1 and M1, respectively. Section 1.5 and 1.5 will then consider attempts to make room for views which answer 'yes' to T1 but 'no' to T2, or 'yes' to M1 but 'no' to M2.

1.2 Propositions

To make sense of T1 and M1, and to explain why they are relevant not just to the investigation of the particular ontological category of *propositions* but to our larger conception of the phenomena of time and modality, we need to get on the table a certain role which all sides can take it for granted that propositions play.

The first and most central part of this role is the *propositional truth schema*:

Truth schema: The proposition that ϕ is such that for it to be true is for it to be the case that ϕ

This is a schema: a type whose instances are sentences. Its instances are sentences like 'The proposition that snow is white is such that for it to be true is for it to be the case that snow is white'. Following standard practice, I will take the class of instances of any schema to be closed under necessitation and universal quantification. Thus, the following will also count as an instance of the Truth Schema: 'Necessarily, for every x , the proposition that x is happy is such that for it to be true is for x to be happy'.

The status claimed on behalf of the sentences that are instances of the propositional truth schema is this: they must express truths, provided that any

ambiguity or context-sensitivity in the sentence substituted for the schematic letter ' ϕ ' is resolved uniformly in both occurrences. This restriction to uniform interpretations is one we standardly take for granted in setting forth schematic logical principles. For example, someone might in a favourable circumstance use the sentence 'Bill Bradley is tall and Bill Bradley is not tall' to say that Bill Bradley is tall for a politician but not tall for a basketball player (which is true). Similarly, under certain circumstances one might use 'If Bill Bradley is tall, he is tall' to say that if Bill Bradley is tall for a politician he is tall for a basketball player (which is false). Recognising this possibility is consistent with endorsement of the schema 'If ϕ then ϕ ' as part of the logic of 'if'. Similarly, we can accept the propositional truth schema while recognising that one could speak truly by uttering 'Bill Bradley is tall, and the proposition that Bill Bradley is tall is not true', or speak falsely by uttering 'If Bill Bradley is tall, the proposition that Bill Bradley is tall is true', provided that one understood the two occurrences of 'tall' non-uniformly.

The other part of the proposition role that I will be taking for granted is sometimes put by saying that propositions are 'objects of belief and assertion'. It turns out that people use 'object' in different ways, but the role I have in mind can be captured by the following schemas:

Belief Schema: The proposition that ϕ is such that to believe it is to believe that ϕ .

Assertion Schema: The proposition that ϕ is such that to assert it is to assert that ϕ .

The general rules given above for interpreting schemas apply here too. So, for example, the belief schema commits us to the claim that necessarily, for every x , the proposition that x is happy is such that to believe it is to believe that x is happy.

'Belief' and 'assert' belong to the category of *attitude verbs*, a category that also includes 'know', 'fear', 'hope', and 'wish'. But for most of these other verbs, the direct analogues of the Belief Schema and the Assertion Schemas cannot be endorsed. For some of the verbs, the problem is that the instances of the schema would be ungrammatical: 'I hope the proposition that snow is white' is ill-formed. For others, the problem is that the instances would be false: there may be some sophisticates who fear the proposition that the Earth is warming (they are "proposition-phobes"), but not everyone who fears that the Earth is warming is like this. I do not take this to count against

the claim that hope and fear are propositional attitudes.⁶ We can capture that claim as follows: there are relations R_1 and R_2 —call them “the hope relation” and “the fear relation”—such that to hope that ϕ is to bear R_1 to the proposition that ϕ , and to fear that ϕ is to bear R_2 to the proposition that ϕ . We must distinguish the fear relation from *fearing*, since one can bear the fear relations to propositions without being fearing them. (Given the belief schema, however, there is no need to distinguish “the belief relation” from *believing*.) Having introduced this useful way of talking, we can use it to state a generalisation of the belief and assertion schemas:

Attitude Schema: The proposition that ϕ is such that to bear the V-ing relation to it is to V that ϕ .

Here V can be replaced by any propositional attitude verb. It is hard to imagine a principled view that accepts the belief and assertion schemas but regards it as impossible to interpret ‘the hope relation’, ‘the fear relation’, etc. in such a way as to make the instances of the attitude schema true. However, we will not need to rely in any essential way on the general attitude schema in what follows.

Are the above schemas as unproblematic as they seem? One reason for worrying is ontological: a certain kind of nominalist will maintain that there are no propositions at all, since there are no abstract entities at all, and thus hold, for example, that the Earth is round although it is not the case that the proposition that the Earth is round is true (since there is no such thing as the proposition that the Earth is round). But philosophers who find views of this sort attractive almost always go on to introduce some piece of ideology which would allow them to make a distinction, among those sentences which they refuse officially to endorse because of their mistaken ontological presuppositions, between “good” sentences (like ‘The proposition that the Earth is round is true’) and “bad” ones (like ‘The proposition that the Earth is trapezoidal is true’). Moreover, there is reason for philosophers of this sort to take the status of ‘goodness’ to be preserved by logically valid arguments. Thus, it may be possible to set nominalistic scruples aside in order to get at the underlying issue about time, simply by explaining that our uses of sentences involving quantification over propositions is to be understood as merely aiming at the relevant form of goodness.

Proposition-talk is useful in branches of metaphysics besides the metaphysics of propositions because it allows us to achieve certain express-

⁶See King 2002, ***versus Merricks***.

ive goals. It is instructive to consider another form of language which would achieve the same goals. In logic, *second-order* languages contain variables which function syntactically like predicate letters—including variables which function syntactically like sentence letters (0-ary predicate letters). These languages thus allow for *quantification into sentence position*. We could imagine supplementing standard English with sentential variables, and the quantifiers that bind them. The grammatical sentences in the new language would include sentences like the following:

- (1) a. $\exists P(\text{necessarily, } P \text{ if and only if the Earth is round}).$
- b. $\exists P(P \text{ and John believes that } P).$
- c. $\exists P(P \text{ and it is not always the case that } P).$

There is a natural “translation” from the sentences in the new language to sentences in the old language, which involves replacing the sentential variables by ordinary objectual variables; replacing each quantifier binding a sentential variable with an ordinary quantifier restricted to propositions; adding ‘is true’ to each variable that is not the argument of some attitude verb; and replacing expressions like ‘hopes that x ’ with ‘bears the hope relation to x ’. Those who hold that such linguistic innovations depend for their intelligibility on some translation back into the original language will have no choice but to rely on some translation along these lines in making sense of quantification into sentence position. However, there is a more tolerant attitude, which allows that some philosophically interesting languages can be learnt ‘by immersion’.⁷ If you take this liberal attitude, and you think you can understand quantification into sentence position, you might find it dialectically helpful to replace claims expressed using quantification over propositions with sentences involving quantification into sentence position whose translations are logically equivalent to those claims. For example, you can substitute (1c) for the propositional temporalist’s claim ‘Some propositions are true but not always true’. If you are tempted to take a different attitude towards the resulting claim from the one you took toward the original claim, that is at least a sign that you should watch out. Perhaps something non-standard about your theory of propositions is preventing your views about propositions from fitting together with your views about the metaphysics of time in the way that would make the former a good way of getting at the latter.

* * *

⁷See, e.g., Williamson 2003, p. 459.

Having established the required background, let us return to T1 and M1. Propositional temporalists hold that just as a person can be tired sometimes without being tired all the time, a proposition can be true sometimes without being true all the time. (Indeed, typical propositional temporalists will think that whenever a person is tired sometimes without being tired all the time, the proposition that that person is tired *is* one of the propositions that is true sometimes without being true all the time.) Propositional eternalists, by contrast, hold that a proposition could no more be temporarily true than a number could be temporarily prime. Similarly, propositional contingentists hold that there are true propositions which could have been false, while propositional necessitarians hold that a proposition could no more be contingently true than a number could be contingently prime.

(The obvious way to express propositional eternalism [necessitism] is to say that each true proposition is always [necessarily] true, just as each prime number is always [necessarily] prime. Temporal [modal] modifiers are thus *inert* when applied to 'prime' and 'true'. However, some people who should clearly count as propositional eternalists and necessitarians might resist these formulations. To get a sense of the worry, consider spatial modifiers like 'somewhere' and 'everywhere'. While one could hold that spatial modifiers apply only inertly to predications of 'prime', so that 'Each prime number is prime everywhere' is true, it is not obvious that this is the right view. 'The number two is prime under my chair' is a very strange sentence; perhaps its strangeness is best explained by denying that it is true. (Compare 'The number two is happily prime' and 'The number two is prime with a knife'.) If we decide to say this, and are prepared follow the B-theorists' guiding analogy between time [modality] and space where it leads, we might be led to say regard 'The number two was prime last Tuesday', and perhaps even 'The number two would have been prime if I had missed the bus this morning', as untrue for the same reason.⁸ And if we get this far and are propositional eternalists [necessitarians], we will presumably want to treat 'true' in the same way as 'prime', leaving us with a view on which sentences like 'Some proposition was true last week' ['Some proposition would have been true if I had missed the bus'] are never true.

⁸Burgess, 'Numbers and Ideas' endorses this thought: 'Now creating a language involves creating certain rules for its use. Among these is, I believe, a rule to the effect that tense and date are not to be applied to mathematical existence assertions. One can say "There exist infinitely many prime numbers," but to ask "How many of them already existed in 1000 BCE, or during the Cenozoic Era?" is to commit a kind of grammatical solecism.'

As it happens, I think that this way of developing propositional eternalism and necessitism is a mistake: the strangeness of the relevant mathematical sentences is better explained by the claim that the relevant temporal and modal modifiers are vacuous. However, it is important not to allow the relatively shallow dispute over the proper treatment of these sentences to obscure the underlying issue.)

How is propositional necessitarianism a coherent position? The following argument for propositional contingentism seems watertight: Obama is president. But it is not necessary that Obama is president. The proposition that Obama is president is such that, necessarily, it is true iff Obama is president. Therefore, the proposition that Obama is president is true, but not necessarily true.

The key premise in the foregoing argument was an instance of the following

Modal truth schema: The proposition that ϕ is such that necessarily: it is true iff ϕ .

A propositional necessitarian could resist the argument by simply giving up on this schema. But given that we are taking the Truth Schema for granted, this looks very unpromising. If we have any understanding at all of claims of the form 'For it to be the case that ... is for it to be the case that ...', we know that they entail the corresponding necessitated biconditionals. If for a certain glass to contain water is for it to contain H₂O molecules, it is necessary that it contains water iff it contains H₂O molecules. This much should be uncontroversial: the live debate is whether there is any interpretation of necessity that validates the reverse inference from 'Necessarily P iff Q' to 'For it to be the case that P is for it to be the case that Q'. But given this, each instance of the truth schema commits us to the corresponding instance of the modal truth schema, and the argument against the propositional necessitarian stands.

There is a parallel argument for propositional temporalism. Obama is president, but he will not always be president. But the proposition that Obama is president is such that it will always be true iff Obama is president. Therefore, the proposition that Obama is president is true, but will not always be true. The key premise in this argument is an instance of the following:

Temporal truth schema: The proposition that ϕ is such that always: it is true iff ϕ

And again, given that one accepts the truth schema, a flat rejection of the temporal truth schema looks problematic. Doesn't 'For it to be the case that P is for it to be the case that Q' entail 'Always, P iff Q'? Even if one doubts the validity of the inference from 'Necessarily P' to 'Always P' (perhaps on the grounds that it is necessarily but not always true that it is raining if and only if it is raining now), these doubts do not extend to claims of the 'For it to be... ' form. (This was my motivation for formulating the truth and attitude schemas using such claims rather than necessitated biconditionals). The claim that to be water is to be H₂O manifestly rules out the suggestion that there will be water that is not H₂O in the future, or that there was any such water in the past.

Propositional temporalists and contingentists should be wary of relying on these arguments. For there is an exactly parallel argument for the (much less popular) claim that propositions can be true in some places and false in others. This argument would appeal to the following

Spatial truth schema: The proposition that ϕ is such that *everywhere*, it is true iff ϕ

Using this, we can argue from the premise that Obama is popular in California and not popular in Arkansas to the conclusion that there is a proposition (namely, the proposition that Obama is popular) which is true in California and not true in Arkansas. Few would be willing to follow this argument where it leads. But the reasons for thinking that the truth schema entails the modal and temporal truth schemas seem to carry over to the spatial case.

Perhaps, after all, propositional eternalists and necessitarians just have to deny the inference from the truth schema to the temporal and modal truth schemas. But there is a more subtle, and I think more satisfying, response which does not require rejection of any of the schemas. As we have observed, propositional eternalists [necessitarians] must posit widespread context-sensitivity in ordinary sentences. Moreover, they have reason to think that the relevant kind of context-sensitivity has an important feature in common with the kind of context-sensitivity introduced by pronouns like 'he' and 'she', a feature I will call *bindability*. By modelling their account of the relevant kind of context-sensitivity on the context-sensitivity of pronouns, propositional eternalists and necessitarians can agree that the instances of the temporal and modal truth schemas are true on all uniform interpretations.

Here is what I mean by 'bindability'. 'He is tall' is obviously a context-sensitive sentence. There is no one proposition that it must always be used

to assert, since we can use it to assert that any (salient, male) person is tall. But it is also context-sensitive in another way. As well as contributing propositions to the effect that this or that person is tall, it can also make a quite different kind of semantic contribution when it is embedded inside certain larger sentences. In 'Every farmer thinks that he is tall', for example, it is natural to interpret the pronoun 'he' as *bound* by 'every farmer'. When we are interpreting the sentence in this way, it is misguided to ask which proposition is expressed by the embedded occurrence of 'He is tall': the role played by the embedded sentence is not that of contributing any single proposition. From the propositional eternalist's point of view, there looks to be a similar contrast between unembedded uses of 'Susan is happy' and the interpretation it naturally bears when embedded under temporal modifiers, e.g. in 'Whenever there is a party, Susan is happy'. Given propositional eternalism, it seems odd to think of this embedded occurrence of 'Susan is happy' as contributing, or bearing any semantically important relation to, some particular proposition, such as the proposition that Susan is happy at the particular time *t* at which the utterance is taking place. Associating such a proposition with the embedded occurrence does not helpfully characterise its role in determining the proposition expressed by the whole sentence.⁹

The dominant view in the semantics of tense takes the analogy between 'Susan is happy' and 'He is tall' very seriously. On this view, some syntactic constituent of 'Susan is happy' (perhaps the present tense morpheme on 'is', or some unvoiced constituent) has the semantic role of a *variable* ranging over times. Such variables are relevantly similar to the pronoun 'he', in being able to make two quite different kinds of semantic contribution: they can be bound by higher quantifiers (like 'always' and 'whenever there is a party'), but they can also be left unbound, in which case they will refer to some particular time in the same way that an unbound occurrence of 'he' refers to some particular person. But whether or not they embrace this particular style of syntax and semantics, propositional eternalists have good reason to think that somehow or other, embedded occurrences of tensed sentences admit distinctive interpretations that do not correspond to any interpretations these sentences can bear when unembedded.

This distinction opens up a new line of response to the arguments from the modal, temporal and spatial truth schemas to propositional contingentism, temporalism, and 'spatialism'. Recall that in accepting any of schema of this sort, we are committed to the truth of its instances *only when repeated*

⁹***Cite Salmon 1986.

context-sensitive constituents are interpreted uniformly. Consider the instance of the temporal truth schema that was appealed to in the argument for temporalism:

- (2) The proposition that Obama is president is such that always, it is true iff Obama is president.

Here the second occurrence of ‘Obama is president’ cries out to be interpreted as bound by ‘always’, since if we do not interpret it like this, we will have to regard ‘always’ as vacuous. And even if there is a semantically legitimate interpretation of (2) on which ‘always’ is vacuous, it is not one we find it easy to access, for reasons that are easily understood.¹⁰ Compare ‘Every farmer thinks that farming comes naturally to him’ with ‘Every farmer is such that farming comes naturally to him’. An unbound reading of ‘him’ as referring to some salient male person is much easier to hear for the former, since in the case of the latter, such a reading would make ‘Every farmer is such that’ completely vacuous. But if we interpret the second occurrence of ‘Obama is president’ in (2) as bound, we will not be interpreting the sentence uniformly. For since the first occurrence of ‘Obama is president’ in (2) is not in the scope of any temporal modifier, it must be understood as contributing some particular proposition, such as the proposition that Obama is president at *t* (where *t* is the time of utterance). The upshot is that while (2) comes out false on its *natural* interpretation, this is no objection to the temporal truth schema, since the interpretation in question is a non-uniform one.

For an analogy, consider the following obviously correct schema (which follows from the belief schema and the truth schema):

- (3) If ϕ , then everyone who believes that ϕ believes something true

It is no objection to (3) that the following instance admits of a false interpretation where the first ‘he’ is used to demonstrate some particular person, and the second is bound by ‘everyone’:

- (4) If he is tall, then everyone who believes that he is tall believes something true.

By the propositional eternalist’s lights, apparent counterexamples to the temporal truth schema like (2) can be dismissed in the same way as (4). (The

^{10***}ADD: discussion of Percus 2000 on the question whether such a reading is even possible.

only relevant difference is that whereas uniform readings of (4) where both 'he's refer to some particular person are easy to access, uniform readings of (2) are either impossible or at least very hard to access because of our bias against interpretations on which certain constituents turn out to be vacuous.) The same move can be used to defeat the inferences from the modal and spatial truth schemas to contingentism and "spatialism".¹¹

The literature on propositional temporalism and eternalism is haunted by the apprehension that the whole issue might turn out to be in some sense a verbal one. For example, Kaplan, in introducing a view which seems to countenance propositions whose truth values change over time, uses scare quotes around 'proposition', in order, as he puts it, to 'reflect my feeling that this is not the traditional notion of a proposition' (Kaplan 1989, n. 28); he contemplates a view which seems to involve positing propositions whose truth value varies over space, and describes the question whether to adopt such a view as 'largely a matter of linguistic engineering' (p. 504). Remarks like this suggests that there might be at least two legitimate possible uses or senses of 'proposition', and perhaps also of 'true', such that T1 receives different answers depending on how we interpret it.

But if we lay down the truth schema as a non-negotiable constraint on the interpretation of 'proposition' and 'true', it is hard to see how this conciliatory view could be maintained. Suppose there are two available interpretations of

¹¹In formal semantics, the usual way to register facts about the binding of variables uses a system of numerical indices: a variable is bound when it is in the scope of a quantifier carrying the same index. In such systems, there is normally nothing to stop indices from being "recycled". For example, if we posit a variable ranging over times in the syntax of 'Obama is president', there is nothing to stop us from attaching the same index number to the unbound occurrence of that variable in the first occurrence of 'Obama is president' in (2), to the quantifier in 'always', and to the bound occurrence of the variable in the second occurrence of 'Obama is president'. *Prima facie*, this seems to go against my claim that a reading on which the first occurrence is unbound and the second occurrence is bound involves a 'non-uniform' resolution of context-sensitivity. For in a system with indices, it is tempting to cash out the relevant notion of uniformity in terms of the sharing of indices.

However, while the recycling of indices is a feature of standard logical notation, it doesn't seem to correspond to any genuine phenomenon in natural language. There is nothing that would mark the difference between utterances of (2) in which the earlier unbound variable shares an index with the later bound variable, and utterances in which it does not. It seems safe, then, to dismiss this kind of arbitrary index-sharing as an artefact stemming from the use of numerical indices, a practice towards which it is in any case reasonable to adopt a somewhat instrumentalistic attitude.

‘proposition’ and ‘true’, both of which obey the truth schema, such that some propositions₁ are temporarily true₁, but no propositions₂ are temporarily true₁. By the two versions of the truth schema, we have the following:

- (5) The proposition₁ that Obama is president is such that for it to be true₁ is for Obama to be president.
- (6) The proposition₂ that Obama is president is such that for it to be true₂ is for Obama to be president.

These jointly entail (7):¹²

- (7) The proposition₁ that Obama is president and the proposition₂ that Obama is president are such that for the former to be true₁ is for the latter to be true₂.

Given that ‘For it to be the case that *P* is for it to be the case that *Q*’ entails ‘Always, *P* iff *Q*’, (7) yields

- (8) The proposition₁ that Obama is president and the proposition₂ that Obama is president are such that always, the former is true₁ iff the latter is true₂.

thereby ruling out the supposed contrast between propositions₁ and propositions₂.

One could resist this argument by appealing to the conception of ‘always’ as a binder. Perhaps the truth of (7) only requires the truth of (8) when we (unnaturally) read the ‘always’ in the latter as vacuous, so as to preserve an unbound interpretation for all the material in its scope. This leaves open the possibility that (8) has a more natural reading on which it is false. But the view of ‘always’ as a binder—or at least, the version of that view on which the only way to achieve a uniform interpretation of a sentence or argument in which some sentence occurs sometimes in the scope of ‘always’ and sometimes outside it is to read ‘always’ as vacuous—is not neutral as between temporalists and eternalists: it effectively presupposes eternalism.

¹²I am assuming here that ‘For it to be the case that *P* is for it to be the case that *Q*’ has the logic of an equivalence relation. This is controversial. Some think of these kinds of claims as entailing something about the ‘priority’ of one side over the other; they will thus deny that ‘For the glass to contain water is for it to contain H₂O’ entails ‘For the glass to contain H₂O is for it to contain water’. But even these philosophers should have no trouble introducing a subsidiary reading of the ‘For it to be the case. . .’ claims on which this inference goes through; they should interpret my claims as involving this notion.

If there were temporarily true propositions, we could use a sentence to express the very same such proposition both inside and outside ‘always’, without ‘always’ being vacuous. Since it is of the essence of the conciliatory view that it is neutral between temporalism and eternalism (regarding each as true on some reading of ‘proposition’ and ‘true’), it is unsatisfactory if the only way to defend it from the foregoing objection involves appealing to a conception of the demands of uniformity that presupposes eternalism.

1.3 Worlds and instants

Let us now turn to questions T2 and M2, and to the task of making explicit the background ontological assumptions about *instants of time* and *possible worlds* which these questions presuppose. In the case of instants of time, the relevant background theory is fairly firmly rooted in ordinary language talk of times as entities. In the case of worlds, on the other hand, the practice is relatively recent and specialised, having become popular among philosophers and linguists mostly as a result of Kripke’s (1963) seminal work on model theory for modal logic.¹³

I will present the relevant roles in the form of a collection of schemas grouped under four broad headings.

(A) Connections between ‘at t' ’/‘at w ’ and other temporal/modal modifiers.

- (F) It will be the case that $\phi \leftrightarrow$ at some future instant, ϕ
- (P) It has been the case that $\phi \leftrightarrow$ at some past instant, ϕ
- (G) It will always be the case that $\phi \leftrightarrow$ at every future instant, ϕ
- (H) It has always been the case that $\phi \leftrightarrow$ at every past instant, ϕ
- (A) It is always the case that $\phi \leftrightarrow$ at every instant, ϕ
- (S) It is sometimes the case that $\phi \leftrightarrow$ at some instant, ϕ
- ($\diamond$) It is possible for it to be the case that $\phi \leftrightarrow$ at some possible world, ϕ
- ($\square$) It is necessarily the case that $\phi \leftrightarrow$ at every possible world, ϕ

¹³It is worth emphasising that Kripke’s *discoveries* were of a mathematical nature: it is far from straightforward to extract from these discoveries an argument for the existence of entities playing the “world” role, let alone an argument that claims about possible worlds are in any sense explanatorily prior to the modal claims to which the role requires them to be equivalent.

(B) Relations between ‘at’ and truth-functional connectives:

(At- $\wedge$) At i , (ϕ and ψ) $\leftrightarrow$ at i , ϕ and at i , ψ

(At- $\vee$) At i , (ϕ or ψ) $\leftrightarrow$ at i , ϕ or at i , ψ

(At- $\neg$) At i , it is not the case that $\phi \leftrightarrow$ it is not the case that at i , ϕ

(C) The *before/after* relation among instants, and the *accessibility* relation among worlds:

(At-F) At t , it will be the case that $\phi \leftrightarrow$ at some instant t' after t , ϕ

(At-P) At t , it has been the case that $\phi \leftrightarrow$ at some instant t' before t , ϕ

(At- $\diamond$) At w , it is possible for it to be the case that $\phi \leftrightarrow$ at some w' accessible from w , ϕ

(At- $\Box$) At w , it is necessarily the case that $\phi \leftrightarrow$ at every w' accessible from w , ϕ

(D) The temporal and modal status of ‘at t ’ and ‘at w ’ facts:

(Permanence) If at t , ϕ then it is always the case that at t , ϕ

(Invariance) If at w , ϕ then it is necessary that at w , ϕ

As before, we count universal closures and necessitations of instances of these schemas as instances in their own right. The claim made in putting forth a schema is that its instances are true provided that all ambiguous and context-sensitive elements are interpreted uniformly.

Now for some commentary.

(i) ‘Possible’ and ‘necessary’ are famously context-sensitive. They admit of a wide range of meanings, which abstract to different degrees and in different ways from how things actually are. Qualifiers like ‘nomic’, ‘metaphysical’, ‘historical’ and ‘practical’ can help to pin down the intended reading, but the context-sensitivity is so extensive that any short list of varieties of necessity and possibility will barely scratch the surface. The intention behind the schemas in group (A) is that ‘possible world’ matches the context-sensitivity in ‘necessary’ and ‘possible’: so if ‘possible’ means ‘nomically possible’, ‘possible world’ means ‘nomically possible world’, and so on. Similarly, ‘accessible’ in group (C) is to be thought of as varying in tandem with ‘possibly’ and ‘necessarily’.

It is less clear whether temporal modifiers like 'it will be the case that' introduce any context-sensitivity. But if they do, then we should interpret expressions like 'future instant' and 'after' as context-sensitive in a parallel way.

(ii) Among the readings of 'possibly' and 'necessarily' are *epistemic* readings: for it to be epistemically necessary that P is, roughly, for relevant people to know, or to be in a position to know, that P . Trying to treat epistemic modality within the framework of worlds puts pressure on several components of the 'world' role. For by repeated application of the principles in group (B), we can derive 'At w , ϕ ' whenever ϕ is a truth of propositional logic. But 'It is epistemically necessary that ϕ ' is not true in all such cases, since (*pace* a long tradition of heroic bullet-biting still carried on in philosophy by the followers of Stalnaker, and in formal semantics by almost everyone) we are not omniscient about logic. One response to this is to give up (B), allowing for metaphysically impossible but epistemically possible worlds which are logically ill-behaved in the relevant ways. While this is not a hopeless endeavour, it brings up many complexities. Given our interests in this book, we will do best to set epistemic modality aside for separate treatment, and stick with the conception of worlds that allows maintaining the schemas in group (B).

(iii) When we know that a certain instant is past or future, it is not very natural in English, although it is not ungrammatical, to use the present tense in talking about how things are at that instant. (Compare 'At the beginning of the new millenium, I was asleep' to 'At the beginning of the new millenium, I am asleep'.) Relatedly, it is hard to hear temporal modifiers inside the scope of 'at t ' in such a way as to fit with $At-F$ and $At-P$. 'At t , it will be the case that I am asleep' sounds equivalent to 'At t , which is in the future, I am asleep', whereas $At-F$ requires us to hear it as equivalent to 'I am asleep at some time after t '. However, the alternative usage needed for my schemas to be tenable, on which past and future tense markers under 'at t ' do not merely introduce presuppositions about the temporal location of t , is not simply a philosopher's invention. It arises naturally when we are quantifying over instants past, present and future: 'At every moment of their waking lives, truly good people strive to make the world a better place'. Or when we don't know, or have reasons not to communicate our knowledge, about the temporal status of a particular instant: 'I am thinking of a certain moment of your life—can you guess which one? I'll give you a clue: at the time I'm

thinking of, you are asleep, but you will be awake within a few seconds.'

(iv) A worry about the temporal schemas which deserves extended discussion concerns certain classes of sentences for which 'at such-and-such instant, ϕ ' sounds extremely odd, prompting the worry that the corresponding instances of any of the schemas might not be true. Three groups of such sentences come to mind.

The first group consists of "atemporal" sentences like 'The number two is prime'; 'I visit Disneyland at least once in my life'; 'World War Two lasts six years'; 'Spacetime is four-dimensional'. Here, I think that by far the best thing to say is that the relevant kind of oddity is just that of *vacuous truth*. In the sentence 'At instant t , World War II lasts six years', the temporal modifier is redundant. But for obvious reasons, we are biased against interpretations on which certain constituents turn out to be obviously vacuous, so the sentence prompts us to flounder around thinking about weird scenarios that might somehow be describable as involving the compression of six years into an instant. By contrast with a view that denies truth in the relevant cases, the vacuity view has the desirable consequence that it lets us regard inferences like 'Next week, my students will know that spacetime is four-dimensional' to 'Next week, spacetime will be four-dimensional' as valid.

The second group involves sentences which, although perhaps temporally variable, characterise things over long periods of time. For example, it sounds odd to say 'At noon on January 1st 2005, I occasionally drink coffee'. In these cases, I think we can get into the swing of understanding the relevant embeddings. You can be an *occasional smoker* at an instant, for example: 'Exactly fifty years, two days and one hour ago, an occasional smoker was asleep in this bed'. Or imagine making a bet: 'At midnight on December 31st, 2050, if I am still an occasional smoker, I will give you a thousand dollars'. Of course, being an occasional smoker at an instant is a matter of what is going on at other nearby instants; but this is so for all sorts of predicates.¹⁴

The third group involves verbs which (intuitively speaking) describe *events* which take a while to happen, as opposed to *states*. When ' t ' stands

¹⁴If our semantic theory uses time variables, we could explain the awkwardness of 'At this instant I smoke occasionally' by saying that 'smoke occasionally' has a free variable that needs to be filled by a longish period of time. An appropriate value for this variable *can* be generated out of 'at t '—e.g. it could be filled in as 'the year surrounding t '—but this does not come naturally to us unless other material in the sentence requires it. Section 2.1 below explains why such a semantics is neutral as regards T1.

for an instant, it is hard to know what to make of 'At t , John kisses Mary'; 'At t , I paint my living room'; 'At t , someone says "the curfew tolls the knell of parting day"', etc.. (Contrast 'At t , John loves Mary' and 'At t , John is kissing Mary'.) The sentences in this group not only sound odd when embedded under 'at t '; they also sound odd when they are asserted unembedded. It is just not clear how to understand 'John kisses Mary' when taken all by itself, out of the blue.¹⁵ One possible family of interpretations involves a *habitual* element, brought out more clearly in 'John kisses Mary, but he never goes any further'. If we can make sense of 'At instant t , John is in the habit of kissing Mary', we should be able to make sense of 'At instant t , John kisses Mary' on this habitual reading. The interesting thing is that no habitual element is required in sentences like 'John will kiss Mary', 'Whenever John kisses Mary, she demurely turns away', etc. Sentences like these do, however, *admit* habitual readings, although they are often rather far-fetched. For example 'Whenever John kisses Mary, Eve kisses Walter' has a hard-to-access reading where it is tantamount to 'Whenever John is in the habit of kissing Mary, Eve is in the habit of kissing Walter'. If we want to accept the schemas above in full generality, we will need to get used to detecting readings of this sort, since they will be required by any uniform interpretations of the schemas under (A).

Sentences in the third group pose a special problem for those A-theorists who endorse the classic tense-logical approach to the analysis of tense in natural language. In this approach, the future and past tense are analysed as operators F and P, applied to sentences that also make sense taken as unembedded. Thus 'John kissed Mary' and 'John will kiss Mary' are analysed as 'P(John kisses Mary)' and 'F(John kisses Mary)', respectively. But what does it take for the embedded sentence 'John kisses Mary' to be true, on the relevant reading? The habitual interpretation is obviously irrelevant. We could say that it is true iff John is in the midst of kissing Mary, or at the first instant of a kiss, or at the last; but each of these options has the bad result that either 'John will kiss Mary' or 'John kissed Mary' or both could be true while John is in the middle of his one and only kiss of Mary. It is much more comfortable to adopt a semantics for the tenses that lets us say something like this: 'John has kissed Mary' is true iff there is an event of

¹⁵Unembedded eventives like 'John kisses Mary' are fine in stories told in the present tense, and in newspaper headlines. Such sentences also have a rather distinctive use in simultaneous commentary on sporting events ('Messi shoots!'). ***ADD: discussion of Ogihara's claim about this.

John kissing Mary that is entirely past, while ‘John will kiss Mary’ is true iff there is an event of John kissing Mary that is entirely future.¹⁶ There is no reason A-theorists cannot adopt a semantics of this sort. Given that propositions can be transient, A-theorists will certainly be able to *make sense* of the operators studied in tense logic. They can, for example, stipulate that whenever ϕ expresses a proposition p , $\lceil \text{F}\phi \rceil$ expresses the proposition that p will be true. But of course this does not commit them to the interpretation of any particular words or morphemes in natural languages as expressing such operators.

(v) $\text{At}\neg$, $\text{At}\vee$ and $\text{At}\neg$ capture the “transparency” of ‘at’ with respect to truth-functional connectives. One can write down similar principles which express the transparency of ‘at’ with respect to the rest of the logical vocabulary of first-order logic, as follows:

(At=) $\text{At } i, y = z \leftrightarrow y = z$

(At $\forall$) $\text{At } i, \text{everything } \phi s \leftrightarrow \text{everything is such that at } i, \text{it } \phi s$

(At $\exists$) $\text{At } i, \text{something } \phi s \leftrightarrow \text{something is such that at } i, \text{it } \phi s$

However, these principles, especially $\text{At}\forall$ and $\text{At}\exists$, are much more controversial than $\text{At}\neg$, $\text{At}\vee$ and $\text{At}\neg$. Given schema $\Box$, acceptance of the modal version of $\text{At}\exists$ would commit one to the extremely controversial principle that everything is necessarily identical to something:

(NE) $\forall x \Box \exists y (y = x)$.

If we follow the logician’s custom of understanding ‘exists’ as synonymous with ‘is identical to something’, this amounts to the claim that *each thing necessarily exists*. Similarly, given H and G , the temporal version of $\text{At}\exists$ would commit us to the claim that each thing always has existed, and always will exist (in the logician’s sense of ‘exist’):

(EE) $\forall x \text{G}\exists y (y = x)$
 $\forall x \text{H}\exists y (y = x)$

¹⁶Some A-theorists (who think of themselves as ‘presentists’) will deny that there *are* any events of John kissing Mary, if John is not kissing Mary. The following will be more widely acceptable: ‘John will kiss Mary’ is true iff there will be an event e such that e is a kissing of Mary by John, and it is not *now* the case that e is or has been happening.

Most modal and temporal A-theorists have rejected NE and EE, on the grounds that many ordinary objects could have failed [will fail, have failed] to be identical to anything at all.

(The labels ‘actualism’ and ‘presentism’ are often understood in such a way as to incorporate a commitment not only to the modal and temporal A-theories, but to the denials of EE and NE. However, official definitions of ‘actualism’ and ‘presentism’ do not always succeed in capturing this commitment, which is part of why I won’t be using this terminology.¹⁷)

(Is there any coherent version of the *B-theory* on which $At-\exists$ and $At-\forall$ fail? Arguably yes: it is open to propositional eternalists [necessitists] to say that all quantifiers introduce a hidden argument place for a time [world], over and above any argument places contributed by the predicates with which they combine. Thus ‘Every philosopher is happy’ might look like ‘Every(t_1) philosopher(t_2) is happy(t_3)’ when these variables are made explicit. However, this view seems to allow one to introduce a non-relativised interpretation of quantification just by binding the variables introduced by quantifiers with an appropriate quantifier of their own: ‘ $\forall t_1$ every(t_1) philosopher(t_2) is happy(t_3)’. And this seems to be a better deserver of the label ‘absolutely unrestricted quantification’ than any of the relativised meanings.¹⁸)

(vi) The desire to find some metaphysically deep truth in the slogan that ‘the future is open’ has motivated some to contemplate giving up various parts of the standard role for instants. One thought is we should capture the “openness” of the future by rejecting $At-\forall$ and $At-\neg$ for future instants. On this picture, where t is an instant one day into the future, and it is “open” whether there will be a sea battle going on one day hence, it is neither the case that at t , a sea battle is being fought, nor that at t , no sea battle is being fought. The view also requires a rejection of Permanence, since although t is currently an ‘open’ instant, it will not always be open. The problem with this it is that it requires giving up the obvious equivalence of ‘not always’ and ‘sometimes not’. Since it is not the case that at t , a sea battle is being

¹⁷Williamson (2010) gives an illuminating critique of the way the expressions ‘actualism’ and ‘possibilism’ have been used in the literature, and argues convincingly that they should be dropped.

¹⁸Even the initial universal quantifier over instants might be thought to require relativisation to some particular instant, in which case we might be able to achieve even greater generality by universally quantifying this too; one might suppose that this generality-increasing can be iterated without limit. But if one accepts the ontology of instants, it is far from clear how a propositional eternalist could coherently think that the facts about which *instants* there are can vary from one instant to another.

fought, it is not always the case that a sea battle is being fought; but for all we have said, there may be no instant at which no sea battle is being fought, so ‘sometimes, no sea battle is being fought’ might be true.

Recent work by enthusiasts for openness (*cite: Cameron, Barnes, Williams; others?*) has tended, instead, to invoke the distinctive ideology of *determinacy* that has been developed in the literature on vagueness. Using this ideology, we can say that it is *indeterminate* whether there will be a sea battle, and thus indeterminate whether there is any future instant at which there is a sea battle, without giving up any of the standard role. On one version of the view, there are future instants such that it is indeterminate what is the case at them. On another version, it always determinate what is the case at a given instant, but some instants are such that it is indeterminate whether they are *in the future* or neither past, present, nor future.¹⁹ On these interpretations, the openness of the future raises no trouble for the standard instant role, and can be entirely ignored so long as we are not ourselves theorising using the ‘determinately’ operator.

(vii) The schemas under (D) are especially important in understanding the issue between elitists and egalitarians. One could imagine other conceptions of entities that satisfy the rest of the world or instant role, but such that the facts about what is the case at one of these entities were subject to change or contingency. For example, we could introduce an extended meaning for ‘at’ that allowed us to talk about how things stand ‘at’ a real number, in accordance with the following rule: where t is any instant and n is any real number, at t , at n , P iff at $t + n$, P , where $t + n$ is the instant n hours after t (if n is positive) or $-n$ hours before t (if n is negative) or t itself (if n is zero). If we count the positive real numbers as ‘future’, the negative ones as ‘past’ and 0 as ‘present’, and interpret ‘before’ as ‘less than’, it is easy to see that taking the real numbers to be the instants in this way preserves (A)-(C) but not (D). (If it isn’t raining but will be raining 17 hours hence, then it isn’t raining at 0, but it is raining at 0 at 17.) We might think of the real numbers in this way of talking as “points in the A-series”; nevertheless, there is nothing to prevent B-theorists from partaking in the practice. And

¹⁹On the latter version of the view, we have to give up the attractive principle that the before/after order is *total* (whenever two instants are distinct, one is before the other). However, we can accept both the restriction of that principle to the case where the two instants are before some third instant, and to the case where the two instants are after some third instant. Among the instants which are such that it is indeterminate whether they are n seconds after the present instant, there is only one which *is* n seconds after the present instant; and definitely. ***ADD: discussion of Belnap’s way of setting this up.

even B-theorists should be happy to agree that the unit class of 0 (the ‘present instant’) is more natural than the unit classes of most other real numbers. This isn’t just because of other numerical properties of 0 that are irrelevant to its role as an instant: the rule that determines how things are at 0 (at a given ordinary instant) is intrinsically simpler than the rules that determine how things are at any other real number (at a given ordinary instant). Thus, if we want the claim that the unit class of the present instant is especially natural to pick out a distinctively A-theoretical commitment, it is important to understand ‘the present’ as denoting an instant on the ordinary conception which includes Permanence. Similarly, giving up the claim that facts about how things are at a world are necessary would leave us with a quite alien conception of worlds, in which the question how natural the unit class of the actual world is relative to those of other worlds might fail to have its expected metaphysical significance.

1.4 Elitism and egalitarianism

Having got the required background theories of propositions, instants, and worlds into focus, let us return to the arguments from section 1.1 to the effect that the answers to T1 and T2, and to M1 and M2, are the same. Since the more controversial direction of those arguments is the one from propositional temporalism to elitism about instants, and from propositional contingentism to elitism about worlds, the present section will focus on that.

Let us call an instant or world *accurate* iff the propositions true at it are exactly the true propositions. The suggestion is that the classification of instants or worlds as accurate or inaccurate “carves at the joints”, so that the class of accurate instants or worlds is highly natural. For propositional temporalists and contingentists, this class thus draws an interesting boundary *within* the class of instants or worlds: one instant or world is accurate, but most instants or worlds (indeed, on plausible assumptions, all but one) are inaccurate. For propositional eternalists and necessitarians, by contrast, all instants or worlds are accurate. They can thus agree that the class of accurate things is highly natural without thinking that the present time, or the actual world, is thereby in any way “distinguished”.

One way of resisting this argument from temporalism [contingentism] to elitism is to deny that the present instant [actual world] is the *only* accurate one. This seems like an unpromising line, for several reasons. (a) It is clear from the temporalist [contingentist] point of view that there are many

inaccurate instants [worlds]. We can use the following template to argue for the existence of inaccurate instants [worlds]: either P or not P ; so either the proposition that not- P is untrue, or the proposition that P is untrue; so either all instants [worlds] at which the proposition that not- P is true are inaccurate, or all instants [worlds] at which the proposition that P is true are inaccurate; but there are instants [worlds] of both sorts; so in either case, some instants are inaccurate. We are thus left with the central thought behind elitism, namely that of a natural division within the category of instants [worlds] between the accurate and the inaccurate ones. (b) In the temporal case, there is the following argument that there is at most one accurate instant. At every instant t , every other instant is either past or future. But no instant is past or future at itself. So for any two instants t_1 and t_2 , there is at least one proposition that is true at t_1 and false at t_2 , namely the proposition that t_2 is either past or future. Hence, if t_1 is accurate, the proposition that t_2 is either past or future is *true*, although it is false at t_2 . In that case, t_2 is not accurate.²⁰ (c) We can make the following appeal to theoretical economy: the point of instant-talk [world-talk] is to fit in a certain way with other kinds of temporal [modal] talk; but if there were two instants [worlds] at which exactly the same propositions were true, this would make no difference to anything else in our temporal [modal] talk; so we might as well keep life simple by adding it to our conception of instants [worlds] that there are no such indiscernible pairs.

A second, and more interesting, strategy for resisting the argument from temporalism [contingentism] to egalitarianism is to say that *no* instants [worlds] are accurate. This is an odd picture, but perhaps it can be made coherent. It might go along with a more general way of thinking about 'at' on which it is quite commonplace for a proposition is true 'at' some part of reality without being true *simpliciter*. For example, one might say that the proposition that *there is no beer*, although false *simpliciter*, is true at my house. (I mean the proposition expressed by 'there is no beer' on an unrestricted interpretation of its quantifier.) However, this view does not fit well with the background theory of instants and worlds laid out in section 1.3. For (a) given the background theory, it is hard to resist the conclusion that every instant and world is accurate *at itself*; and (b) given this conclusion, it is hard to sustain the claim that no instant or world is accurate. As regards (a):

²⁰One might resist the assumption that it is *necessary* that no instant is past or future at itself, on the grounds that time could be circular; but the argument does not have to be considered as an attempt to establish necessary truth.

consider an arbitrary instant or world i . By (D), a proposition p is true at i iff it is true at i at i . So for any p and i , at i (p is true iff p is true at i). If we assume that the facts about *which propositions there are* are the same at all instants and worlds, it follows that at i (for any proposition p , p is true iff p is true at i); in other words, that at i , i is accurate. (The assumption that the facts about which propositions there are do not change is controversial, but the ways in which it has been supposed to fail would be of no help in trying to sustain a view on which some instants fail to be accurate at themselves.) As regards (b): it follows from (a) that there is *always* an accurate instant, and that there is *necessarily* an accurate world. The proposition that no instant [world] is accurate is always [necessarily] false. Thus, whenever anyone believes that no instant is accurate, they believe something false. And necessarily, anyone who believes that no world is accurate believes something false. And it seems utterly incoherent to concede that this is so, while insisting that nevertheless no instant is accurate. How could one rationally be indifferent to the fact that one believes something which always, or necessarily, involves believing a falsehood?

A third strategy for resisting the argument is to concede that there is a unique accurate instant [world], while denying that its unit class is especially natural. The proponent of this strategy thus thinks of *accuracy* as relevantly similar to properties like *being John Malkovich* or *being the instant 8,124,309,463 years after the beginning of time*. John Malkovich is the only person who enjoys the distinction of *being John Malkovich*, but his unit class is not especially natural compared with those of other people. The reason is that *being John Malkovich* belongs to a large class of roughly equally natural properties, namely those of the form *being identical to x* , where x is a person. Similarly, *being the instant 8,124,309,463 years after the beginning of time* belongs to a large class of roughly equally natural properties, namely those of the form *being n years after the beginning of time*. But how could it make sense to think of *accuracy* like this, as a member of some large family of roughly equally natural properties? What would the relevant family be? In the modal case, it is hard even to think of candidates. In the temporal case, we might consider other properties of the forms *being an instant such that the propositions true at it are exactly those that will be true in n seconds time* and *being an instant such that the propositions true at it are exactly those that were true n seconds ago*. But if we go by the forms of the predicates we need to use to express these in our language, these properties seem considerably less natural than accuracy. To express them, we need this extra material about 'in n seconds time' or ' n seconds

ago', which corresponds to nothing in our definition of accuracy. One could try to think of this as a mere idiosyncrasy of our language. Might there be other languages, "metaphysically on a par" with ours, in which the property of *being going to be an accurate in an hour* is as easy to express as *accuracy* is for us, and in which *accuracy* is as hard to express as *having been accurate one hour past* is for us? But these languages do not *seem* to be metaphysically on a par—it is far from being a mere coincidence that we speak the kind of language that we actually speak, rather than one of these alternatives. It would surely be objectively extremely unlikely that any community should end up speaking one of those languages. And that is not just because of some idiosyncratic feature of the actual laws of physics, which privileges communities like ours over communities of that sort: any laws of physics under which the evolution of such languages was objectively likely would have to be ridiculously fine-tuned and gerrymandered by comparison with the actual laws.

We do not need to lean on the property of accuracy itself in order to make this kind of argument. Take the property of *being spherical*, for example. This is intuitively fairly natural (non-gerrymandered); the fact that we have a predicate that expresses this property seems far from being an arbitrary or idiosyncratic feature of our language. The two-place relation *being an x and t such that x is spherical at t* also seems rather natural. In terms of these, we can define the property of being "accurate as regards sphericity": *being an instant [world] such that all and only the spherical things are spherical at one*. To sustain the view that the unit class of the present instant [actual world] is not especially natural, we would need to say the same thing about *accuracy as regards sphericity* that we said about accuracy itself: that despite appearances, it is no more natural than other properties (e.g. *being such that one will be accurate as regards sphericity in 17 hours time*) that single out other instants [worlds]. If this strategy for preserving egalitarianism is to work at all, it looks like its proponents will have to deny, for example, that the class of spherical things is any more natural than any of the other classes that sometimes [possibly] comprise all and only the spherical things. More generally, they will be pushed towards the claim that only the eternal [necessary] properties of things are relevant to the facts about the relative naturalness of classes.

My main worry about this way of preserving egalitarianism is that it deploys the word 'natural' in a way that severs central connections between naturalness and other notions, connections without which there is no hope

of bringing any discipline to debates about naturalness.²¹ I have already mentioned one of these connections, namely the idea that natural properties are ‘reference magnets’—properties that one can easily express, or have as a semantic value. But I could equally well have appealed to the connection between naturalness and *similarity*, a connection which Lewis emphasised in his initial introduction of the concept ‘natural’. To the extent that a property (or class) is natural, items that share it tend to be similar, while items that are divided by it tend to be dissimilar.²² Given any workable conception of this connection, it looks like the facts about which classes are natural need to be changeable [contingent] if the facts about similarity and dissimilarity are changeable [contingent]. And from the point of view of a propositional temporalist [contingentist], it looks quite obvious that the domain of changeable [contingent] propositions should be taken to include propositions about similarity and dissimilarity. For example, the currently spherical things resemble one another quite a lot right now, but will resemble much less later, after some of them have changed shape.

A third important aspect of the naturalness role is the idea that truths about the instantiation of natural properties and relations constitute a *supervenience base* for all other truths.²³ This makes trouble for the following combination of views: (a) there are no very natural transient properties or relations; (b) necessary truth entails eternal truth; (c) there are temporarily true propositions. For by (b), the temporarily true propositions cannot supervene on any body of eternal truths, and by (a) the truths about the instantiation of very natural properties and relations are eternal. One could resist this argument by denying (b), holding for example that the transient proposition that Obama is president is *necessarily* equivalent to the eternal proposition, concerning the present instant, that Obama is president at it (see Dorr and Goodman n.d.). But one might worry that if there can be necessarily but not

²¹For an attempt to lay out these connections systematically, see Dorr and Hawthorne 2013.

²²This formulation is intentionally vague and non-committal. One way of making it more precise would be to think of the degree of dissimilarity between two objects as something like the sum of the total degrees of naturalness of the properties (or classes) that divide them, where this is measured in such a way as to allow this sum to converge—see Dorr and Hawthorne 2013, p. ? Of course words like ‘similar’ are highly context-sensitive; for some ideas about how this could be taken into account in articulating a connection between similarity and naturalness see Dorr and Hawthorne 2013, p. ?

²³For Lewis, even the *perfectly* natural properties constitute such a supervenience base; but it follows from this that all of the *reasonably* natural properties do so.

eternally true propositions, it is not clear that the merely modal claim that the natural properties constitute a *supervenience* base for all truths really does a good job of articulating the underlying thought. As we saw in section 1.2, this combination of views makes it tempting to think that many philosophically interesting ideas customarily expressed in terms of necessity really need to be expressed in some other terminology, so as to entail not just the relevant claims about how things necessarily are but also the corresponding temporal claims. For example, 'Necessarily, one knows that P iff one has a safe true belief that P ' seems too weak to capture the distinctive thought behind the safety analysis of knowledge, if it is compatible with the claim that it will soon be the case that some people know that P while lacking safe true belief; we would do better, arguably, to replace the modal claim with the claim that *to know that P is to have a safe true belief that P* . Arguably, those who deny that necessity entails eternity have the same reason to replace the idea that the natural properties constitute a supervenience base for everything with some stronger claim in the language of "identifications"—perhaps the claim that for every proposition, there is a truth (perhaps infinitary) of the form 'For this proposition to be true is for it to be the case that ϕ ', where ϕ is expressed in a language all of whose predicates express natural properties. Even if this isn't quite right, the worry that the strategy of resisting the argument by denying (b) simply exploits a flaw in a particular supervenience-based formulation of the relevant premise is quite a pressing one.

The terminology of relative naturalness is the clearest and most disciplined device we have for regimenting the kinds of questions that also get formulated using expressions like 'privileged', 'distinguished', 'metaphysically special' and 'on a par'. But the terminology achieves this discipline because of the array of connections which philosophers theorising about naturalness have forged between 'natural' and other vocabulary. I don't want to claim that any of these connections has the status of an analytic truth: each can be profitably debated, provided certain other things are held fixed. But if too many connections are given up simultaneously, the risk is that debates about naturalness will start to exhibit the kind of pointlessness people sometimes have in mind when they talk about debates being 'merely verbal'. We begin to lose our sense that the different claims being made about naturalness reflect important differences between metaphysical world-views, as opposed to trifling differences as regards the proper application of some philosophical term of art. I don't want to suggest that this is enough to put paid once and for all to the idea that we could co-

herently combine propositional temporalism with temporal egalitarianism, or propositional contingentism with modal egalitarianism. But I do want to suggest that the defenders of such hybrid views bear a heavy explanatory burden, of saying something more about the difference between the realms of the temporary [contingent] and the eternal [necessary], which could serve to justify and make intelligible the claimed contrast regarding naturalness.

1.5 Non-factualism and Relativism

As we saw in the previous section, it is hard for propositional temporalists [contingentists] to resist agreeing that one instant [world] is unique in being accurate. Given that accuracy seems to have the hallmarks of a natural property—especially, that of being *easy to refer to*—this yields a *prima facie* case that propositional temporalists [contingentists] should be temporal [modal] elitists. To resist this *prima facie* case, temporalists [contingentists] need to say something more about the status of temporary [contingent] propositions, something which could make the denial that they need to be taken into account in making determinations of naturalness look like a principled move (as opposed to an arbitrary decision to use ‘natural’ in an idiosyncratic way). In this section, I will consider a proposal which would respond to this need: the proposal that temporary [contingent] questions, such as the question which instant [world] is actual, are *non-factual* questions, and for this reason irrelevant to inquiry into the joints in nature.

To introduce the notion of non-factuality that this suggestion requires us to make sense of, let us consider the history of expressivism in ethics. At the outset, expressivists (most famously, Ayer 1936) went around saying radical-sounding things like ‘Ethical sentences are not true or false; they do not express propositions, but rather perform the quite different linguistic function of expressing non-cognitive mental states’. But then it started to strike them that there must surely be *some* way of interpreting claims like ‘The proposition that lying is always wrong is true iff lying is always wrong’ and ‘The proposition that lying is always wrong is believed by some Kantians’ in such a way as to make them true. Thus was born the tradition of “quasi-realism” (Blackburn 1993, Gibbard 1990), which purported to reconcile expressivism with such ordinary claims. For example, quasi-realists gave semantic accounts of ‘true’, ‘proposition’ and ‘believes’ that allowed ‘The proposition that lying is always wrong is true’ to have exactly the same

attitude-expressing semantic profile as ‘Lying is always wrong’, and allowed ‘Kantians believe that lying is always wrong’ to be semantically equivalent to ‘Kantians disapprove of all cases of lying’ (or some other such attitude ascription).

Once this had been done, however, it was no longer clear that there was anything left to the idea of expressivism conceived of as an alternative to a standard “realist” conception the metaphysics of ethics. Quasi-realists could still endorse a rather distinctive array of claims in the philosophy of mind, e.g. that the state of *believing the proposition that lying is wrong* is identical to the state of *disapproving of lying*. But these claims seem to be consistent with *any* answer to the metaphysical question what it *is* for lying to be wrong, even the most stereotypically “realist” answers of the sort that are arguably deeply at odds with a naturalistic picture of the world. This is what Dreier (2004) dubs ‘the problem of creeping minimalism’.

Some have concluded from this history that the whole appearance that there was a view in the vicinity of expressivism that was neither obviously wrong nor merely a contribution to the philosophy of mind was simply a mirage. Others, however, have drawn the opposite conclusion: that since there *is* an intelligible issue as regards whether ethical questions are “factual” or not, the coherence of quasi-realism just goes to show that a negative answer to this question is consistent with various claims that might have been thought to characterise “realism”—that there are ethical propositions which are true or false, whose truth value does not counterfactually depend on our tastes or practices, which are believed by some and disbelieved by others, etc. To get the deep issue onto the table, we need to invoke some distinctive, high-powered metaphysical vocabulary, which does not admit of any straightforward explanation in terms of the ordinary notions (like ‘proposition’ and ‘true’) to which the quasi-realist treatment applies. For example, we could make a distinction between the ordinary words ‘proposition’ and ‘true’ and the special-purpose metaphysical expressions ‘PROPOSITION’ and ‘TRUE’. In these terms, “factualists” about ethics will assert, and “non-factualists” will deny, that the proposition that lying is wrong is a PROPOSITION (and either TRUE or FALSE). Alternatively, our special-purpose vocabulary could be an operator pronounced ‘it is factually the case that’ (or ‘it is objectively the case that’, or ‘it is really the case that’, or ‘it is metaphysically the case that’...): then we will think of factualists about ethics as claiming, and of non-factualists as denying, that either it is factually the case that lying is always wrong, or factually the case that it is

not always wrong (Fine 2001).²⁴

If we can make sense of this sort of ideology, we can use it to answer the challenge which the previous section posed for the combination of propositional temporalism [contingentism] with temporal [modal] egalitarianism. Suppose that one holds that while some propositions are temporarily true, no PROPOSITIONS are temporarily TRUE. Or suppose that one holds that whenever it is factually the case that *P*, it is always factually the case that *P*.²⁵ Either of these claims would, arguably, provide a principled basis for denying that accuracy (or any other temporary property) is natural, despite being very easy to refer to, relevant to similarity and dissimilarity, etc. For insofar as it is supposed to play an important role in metaphysics, the question which classes are natural is naturally classified as a factual question.²⁶

There is great variation among philosophers in their tolerance of such high-powered ideology. Kit Fine, for example, has suggested that the central task of “critical metaphysics” is to investigate questions expressed in terms of special operators like ‘factually’, ‘really’ and ‘grounding’, none of which seem to have much life outside of philosophy (Fine 2009). Among questions

²⁴One choice point for those who theorise in terms of something like a ‘factually’ operator is whether to regard it as *factive* in the sense that ‘factually ϕ ’ always entails ϕ . If one rejects this inference, one can also consider a kind of ‘super-factualism’ about ethics according to which it is both factually the case that lying is always wrong, and factually the case that it isn’t. The ‘fragmentalist’ view defended by Fine (2005c, *check*) makes a move of this sort, but using the operator Fine pronounces ‘really’ rather than the one he pronounces ‘factually’.

²⁵One way to develop the view that truths about what is factually the case are eternal is to say that whenever it is sometimes the case that *P* and sometimes the case that not-*P*, it is never the case that factually *P* or that factually not-*P*. This is close to what Fine (2005c) calls ‘anti-realism about tense’, except that it uses a ‘factually’ operator where Fine’s anti-realist uses a ‘really’ operator. (The differences between Fine’s ‘really’ and ‘factually’ operators make little difference for my present purposes.) Another way to develop the view, available to those who reject the factivity of ‘factually’, is to hold that for certain *P* such that sometimes *P* and sometimes not-*P*, it is *both* factually the case that *P* and factually the case that *P*. Fine preferred view, ‘fragmentalism’, seems to work like this, modulo the substitution of ‘really’ for ‘factually’.

²⁶Alternatively, those who theorise using the ideology of PROPOSITIONS or factuality could *accept* that the unit class of the present instant [actual world] is more natural than the unit classes of most times [worlds], while denying that it is factually the case that it is more natural than them, or that the PROPOSITION that it is more natural than them is TRUE. It is not clear that there is any non-verbal disagreement between propositional temporalists [contingentists] who embrace temporal [modal] elitism while qualifying it in this way, and those who reject temporal [modal] elitism on the grounds that what is factually the case is eternally [necessarily] the case and the only natural properties are those whose instantiation is a factual matter.

in which no such vocabulary figures, the ones which might initially seem to be of interest to metaphysics typically have easy answers. For example, it is supposed to be obvious that there are holes, but not obvious whether there *really* are holes; it is supposed to be obvious that there are temporarily true propositions, but not obvious whether there are temporarily *factually* true propositions.²⁷ At the other extreme, there are many philosophers who find all this ideology utterly unintelligible and disreputable.²⁸

To wax autobiographical for a moment, I find myself somewhere in between these positions. In ethics, the consensus has been that non-factualists have to do a lot of *hard work* to make their view tenable. They have to, at a minimum, formulate a systematic recipe which provides, for each sentence ϕ involving any ethical vocabulary, an answer to the question 'What is it to believe that ϕ ?' in which no ethical vocabulary occurs inside any attitude verb (see Schroeder 2008). While I am not confident that I understand what significance such a theory would have for metaphysics as opposed to the philosophy of mind, I am willing to listen. But if someone claims to be a non-factualist about ethics but feels no special pressure to go in for this kind of ambitious reconstructive project, I begin to feel I have lost my grip on the issue altogether. In domains other than ethics, I have some sense of how non-factualism could be a metaphysically relevant option when I can dimly imagine analogous reconstructive projects, in which the psychological states we normally describe as involving attitudes towards propositions partly about the relevant subject matter are systematically redescribed, in an allegedly more metaphysically perspicuous way, in a psychological vocabulary that makes no appeal to propositions about the subject matter in question.

Can non-factualism about the transient or the contingent be assimilated to this model? What, according to a non-factualist, could be the metaphysically perspicuous description of the psychological state of believing that *P*, when the proposition that *P* is transient or contingent? Perhaps this the thought could be as simple as: what it is to believe, at *t*, the transient proposition that *Obama is president* is to believe at *t* the eternal proposition that *Obama is president at t*, perhaps under a certain distinctive present-tense mode of presentation.²⁹ Thus the very same psychological state can be characterised

²⁷See Fine 2005c, p. 275–6. Using 'fact' in a way that corresponds to 'true proposition' rather than 'factually true proposition', Fine says that 'the claim that there are tensed facts. . . is not something that can be sensibly denied'.

²⁸Hofweber n.d., Lewis 2005; **DworkinOT**

²⁹Alternatively, it might be to believe an eternal Fregean proposition that "builds in" that

either as an attitude towards a temporary proposition or as an attitude towards an eternal one; the non-factualist will add that the latter description is more ‘metaphysically perspicuous’ than the former, in the same sense in which a quasi-realist might think that the description of the state of believing that lying is wrong as *disapproval of lying* is more metaphysically perspicuous. Non-factualism about the contingent could be developed in a similar way. Are these sketches enough to raise non-factualism about the transient and the contingent to the status of debatable options? I am really not sure. It seems suspiciously *easy* by comparison with the hard work that has been thought necessary for the development of expressivism about ethics; but I don’t know how to turn this suspicion into an argument.

Even if the relevant non-factualisms are coherent, it is not clear what would motivate them. In general, we should expect metaphysical questions about what natural joints there are in reality to be of great relevance to linguistic questions about the extent of context-sensitivity in language. For example, the discovery that spacetime is relativistic, so that there is no metaphysically privileged standard of simultaneity, strongly supports the view that sentences of the form ‘Such-and-such happened simultaneously with so-and-so’ are systematically context-sensitive.³⁰ In this case, non-factualism is less plausible than contextualism. It seems very odd to suppose that people who apply ‘simultaneous’ differently from us, perhaps because they are moving with respect to us, thereby end up believing and asserting a lot of false propositions about which things are simultaneous with which. And it’s still odd even if we add that the false propositions in question are not *factually* false. Thus, it looks like the *default* option for temporal [modal] egalitarians is to be propositional eternalists [necessitarians], and hence contextualists about tense [modality]. One would need some compelling reason, perhaps in the form of linguistic data that cannot be accounted for in a contextualist framework, to justify overriding this presumption. Chapter 2 will consider some considerations that might be thought to motivate this position, and find them wanting.

In the literature, many of the proponents of propositional temporalism say things that make it sound like they are taking egalitarianism about time

mode of presentation. The idea of a “present-tense mode of presentation” will be explored further in section 2.5.

³⁰I will have a lot more to say about the philosophical significance of relativity in sections 2.7 and 6.5 below.

for granted, without deploying anything like Fine's high-powered "factually" operator, or engaging in any of the hard reconstructive work one might expect if they were really trying to defend something analogous to expressivism in ethics. Kaplan (1989) is a typical case (*quote*). A slogan endorsed by many of these philosophers is that *truth for propositions is relative to a time*. But the meaning of this slogan, and its bearing on the argument from temporalism to elitism, is deeply obscure.

Consider how we normally take claims of the form '*F*-ness is relative to a *G*', for instance 'Tallness is relative to a contrast class', 'Simultaneity is relative to a frame of reference', 'Provability is relative to a deductive system', etc. We might think that '*F*-ness is relative to a *G*' just means 'Something can be *F* relative to one *G* and not relative to another'. Understood in this way, 'Truth is relative to a time' is nothing more than a statement of temporalism about propositions. However, '*F*-ness is relative to a *G*' is normally understood to convey more than this. The Earth is flat according to some stories and not according to others, but we do not say on that account that 'Flatness is relative to a story'.³¹ The further claim made by '*F*-ness is relative to a *G*' is that the truth about which things are *F* relative to which *G*s in some sense *exhausts* the truth about *F*-ness: once it is settled which things are *F* relative to which *G*s, there is no further open question that about which things are "just *F*". This claim takes us very close to a form of contextualism about the predicate '*F*'. For, in the usual cases, it would be implausible to think that there is one particular *G* such that whenever someone asks 'Is this *F*?' in English, they express a question necessarily equivalent to the question whether the object in question is *F* relative to that particular *G*. Rather, on different occasions when '*F*' is used, the propositions expressed will concern *F*-ness relevant to different *G*s.

On this natural interpretation, then, 'Truth is relative to a time' would amount to an endorsement of a context-sensitive account of 'true', according to which '*p* is true' can be used, literally, to assert many different things, each of which is necessarily equivalent to the proposition that *p* is true *at* (or "relative to") some particular time. But this claim of context-sensitivity is deeply problematic. Let us suppose that a certain context-insensitive sentence ϕ semantically expresses a transient proposition *p*. For the contextualist about 'true', 'The proposition that ϕ is true' is a context-sensitive sentence, which

³¹And I don't think there is much insight to be gained by bearing down hard on the difference between 'according to' and 'relative to'—in these discussions, 'relative to' generally serves as a sort of proxy for many different prepositional phrases.

can be used to assert any of a wide range of propositions to the effect that p is true at this or that time. Since the facts about how things are at a time do not change, all of *those* propositions are eternally true or eternally false, in sharp contrast to p itself. But this drastic semantic divergence between ϕ and 'The proposition that ϕ is true' is highly implausible on its face. As has often been noticed, 'The proposition that ϕ is true' seems like little more than a wordy stylistic variant of ϕ ; assigning the two sentences entirely different semantic profiles fails to respect this close relation. Moreover, the propositional truth schema, which we are taking for granted as a non-negotiable component of the "proposition role", effectively rules out this form of context-sensitivity in 'true'. For if 'true' were context-sensitive, the instances of the schema would have to be context-sensitive in the same way. As uttered at t , 'The proposition that ϕ is such that for it to be true is for it to be the case that ϕ ' will express the proposition *that the proposition that ϕ is such that for it to be true-at- t is for it to be the case that ϕ* (assuming that there is no other relevant context-sensitivity). For the instances of the schema to be true no matter when they are uttered, it would thus have to be the case that for any instants t and t' , the proposition that ϕ has both of the following features: for it to be true-at- t is for it to be the case that ϕ , and for it to be true-at- t' is for it to be the case that ϕ . This entails that for it to be true-at- t is for it to be true-at- t' , which defeats the whole point of the postulated context-sensitivity in 'true'.

(Of course, philosophers are free to stipulate a new meaning for 'temporal proposition' on which it applies to, e.g., properties of instants, or functions from instants to propositions proper. And they are free to stipulate some appropriate meaning for 'true at' when applied to such entities: e.g. ' p is true at t ' could mean ' t has p ' or ' $p(t)$ is true'. They can even stipulate a new context-sensitive meaning for 'true' which allows the hidden argument place to be filled in contextually. However, these are misguided terminological choices, insofar as they are apt to lead those who lose track of the difference between the stipulated meanings and the originals to think that propositional temporalism is trivially true.)

These days, sophisticated philosophers who find something to like in slogans of the form 'Truth for propositions is relative to a G ' take pains to make it clear that they are *not* proposing contextualism about the ordinary predicate 'true' that features in the propositional truth schema.³² While

³²See for example MacFarlane 2005, MacFarlane 2007, Lasersohn 2005, Richard 2008, and Kölbel 2002, and for discussion of the crucial significance of the present point, Cappelen and Hawthorne 2009, chapter 1.

these “relativists” use a polyadic ‘true relative to’ predicate in their semantic theorising, they also recognise a monadic predicate ‘true’ (sometimes called the ‘object language truth predicate’) which plays a quite different semantic role. Consider the relativist about ‘tasty’. According to the relativist, the proposition *that Marmite is tasty* is true relative to some “standards of taste”, and false relative to others. In this regard it is quite unlike the proposition *that Marmite is tasty relative to such-and-such standard*, which is true relative to all standards or none. The proposition *that the proposition that Marmite is tasty is true* works just like the proposition *that Marmite is tasty*: it is true relative to some standards, false relative to others. (Perhaps they are in fact one and the same proposition) By contrast, the proposition *that the proposition that Marmite is tasty is true relative to such-and-such standard* is true relative to all standards or none. Other versions of relativism invoke other entities in place of “standards of taste”. For example, in Egan’s version of relativism (Egan 2007) the key theoretical notion is truth *relative to a person and a time*.

All of this is exactly isomorphic to the standard propositional temporalist account of instants of time, and the standard propositional contingentist account of possible worlds. As we have seen, in those accounts we can (given some plausible further assumptions) pick out the present instant [actual world] as the unique *accurate* one—the one such that whatever is true at it is true. A relativist about ‘tasty’ can, analogously, characterise the accurate standard of taste as the one such that whatever is true relative to it is (just) true. (Similarly, if they speak of truth relative to people at times, they could characterise the accurate ⟨person, time⟩ pairs as those such that whatever is true relative to them is true.) *Prima facie*, this looks to require an “elitist” view of the relevant entities, according to which the accurate one is thereby singled out in a *natural* way. However, real-life relativists think of their view as egalitarian in spirit. The relativist about ‘tasty’, for example, is supposed to claim that (in some sense) there are many “equally good” standards of taste, by contrast with the objectivist who holds that one standard is “privileged”. Unfortunately, as far as I can see, relativists have said almost nothing that could help to *explain* the sense in which they think that all the different Gs are on a par, or how this parity can be reconciled with the claim that only one of them is accurate in the sense defined above.

Thus there is little in the literature on relativism that could help us to understand how propositional temporalism [contingentism] could coherently be combined with temporal [modal] egalitarianism. Relativists do, however, say many things that are apt to engender illusions of understanding on this

issue. Some examples:

- (i) The unfortunate expression ‘truth is relative to Gs’, and the even murkier expression ‘to relativise truth to Gs’, both encourage the thought that relativist view must have something important in common with contextualism in other domains, without doing anything to elucidate what that common thing could be.
- (ii) Some relativists have (with varying degrees of seriousness) suggested certain reductive identifications of propositions with entities belonging to other categories. For example, Egan (2007) suggests identifying propositions with “centred propositions”—roughly, functions from $\langle \text{person, time, possible world} \rangle$ ordered triples to truth values. For example, the proposition that Marmite is tasty might be the function that maps $\langle x, t, w \rangle$ to Truth iff someone similar to x at t at w , but idealised in certain ways, would like Marmite. The surprising thought is that these functions can themselves be true, in the monadic sense of ‘true’ that features in the truth schema. Now, metaphysical egalitarianism about *people* is so obvious that it almost goes without saying.³³ There is no very natural division within the set of $\langle \text{person, time, world} \rangle$ triples which might reasonably be thought to coincide with the distinction between the *accurate* triples—those such that every function mapping them onto the True is itself true—and the rest. But it is the very obviousness of egalitarianism about people that makes it seem so bizarre to think that the items Egan identifies with propositions could be monadically true or false. The obvious and familiar character of the relevant egalitarian claim does nothing to help understand how it can be reconciled with the existence of a distinction explained in terms of monadic truth.
- (iii) Many relativists are committed to the following epistemological claim: each of us ought, at each time, to believe that the true propositions are exactly the propositions true relative to ourselves at that time (or relative to the standard of taste that is “ours” at that time). In other words, each of us should believe him- or herself to be accurate, or to have an accurate standard of taste. Since it is hard to believe that we could be epistemically obliged (even after we have shared all shareable evidence) to hold conflicting views about the location of some natural

³³*cite Prior, Hare, Hellie, Merlo. . . .

boundaries, this epistemological claim creates pressure to think that accuracy somehow fails to draw any such boundary. However, it does nothing to explain how this could be so.

- (iv) Some relativists (**cite!**) have entertained a radical claim in the philosophy of mind that we all necessarily *do* believe ourselves, or our standards, to be accurate—to believe that *P just is* to believe that it is true from one's own perspective that *P*. (On this view, it is impossible not to meet the epistemic obligation in (iii).) Like almost all claims to the effect that certain combinations of beliefs are impossible, this claim is implausible.³⁴ But even if it were true, it is not clear how it would help us understand the relativist's egalitarian claim.³⁵
- (v) Many relativists have emphasised the idea that when one person believes a proposition about the relevant subject matter (e.g. that Marmite is tasty) and another person believes its negation, neither party need be *at fault* (in spite of the fact that, by the propositional truth schema, one of them must believe something that is not true). Of course, this is useless if we take 'at fault' in any epistemological sense, since it is almost uncontroversial that people can come to believe contradictory things while reasoning perfectly from their evidence. But even many non-relativists have thought that there is some other, more objective, but still normative notion in the vicinity, one invoked by slogans like 'truth is the goal of belief'. In these terms, the relativist idea is that truth is *not* the goal of belief—the goal, rather, is truth-from-one's-own-perspective. I am not sure I even have the normative concepts required to enter into this debate. But even if there is a way to make sense of this debate, it is hard to see how downgrading the *normative* importance of monadic truth (and notions like accuracy that are explained in terms of it) would in any relevant way detract from their *metaphysical* status.

³⁴Williamson (2007, chapter 4) makes a sustained case against a wide variety of such impossibility claims.

³⁵It *might* help a bit, by enabling us to see the relativist as making the same kind of claim as the expressivist. On this interpretation, the key idea is that the state of believing that *P* is somehow *more perspicuously* described as the state of believing that it is true from one's own perspective that *P*. However, to carry out this kind of program we would need to find a way to reconstruct *all* attitudes towards perspectivally variable propositions in terms of attitudes towards ordinary propositions. It is not obvious how this would go: for example, the claim that to *assert* that *P* is to *assert* that it is true from one's own perspective that *P* is not one that relativists are likely to find plausible.

What emerges, then, is that relativist's distinctive ideology does nothing to advance the project of understanding how one could coherently combine temporal or modal egalitarianism with temporalism or contingentism. If we are happy to use some special-purpose operator in our theorising without further explanation—whether we pronounce it 'objectively' or 'absolutely' or 'factually' makes little difference—we can use it to explain what our egalitarianism amounts to. But the apparatus of truth-relative-to-a-G does not itself do anything to convey the needed kind of understanding of the relevant operator.³⁶

1.6 Other A-theoretic themes

The dispute between A-theorists and B-theorists about time has been going on for over a century. In that time, a multitude of definitions have been given of the two positions, not obviously equivalent to one another or to any combinations of answers to my questions T1 and T2. It is not my purpose in this book to provide an overview of this body of literature. I hope it is fairly plain from the collection of quotes on p. ?? that my questions are at least *relevant* to the historically important debate, that a certain kind of paradigmatic A-theorist would give a positive answer to both, and that a certain kind of paradigmatic B-theorist would give a negative answer to both. I am not going to argue that my questions exhaust what people have been disagreeing about, or that they are the only intelligible questions in the vicinity. However, I do have a reason for spending a bit of time considering whether there is any coherent and defensible further commitment, not entailed by the combination of T1 and T2, that might reasonably be regarded as integral to the A-theoretic vision. The reason is that I would like to persuade people who are already aligned with the A-theoretic camp to take seriously the distinctive theory I will be developing in the later parts of this book. This theory includes what I think of as the central claims of the A-theory, namely propositional temporalism and temporal elitism; but I predict that some A-theorists will nevertheless feel that it isn't giving them what they really wanted. So, let's consider some of the ideas that often come up in

³⁶Of course, if the relativist is right, it might be the case that 'Objectively ϕ ' and 'Always ϕ ' are always necessarily equivalent, in which case our formal semantics could reflect this by assigning exactly the same semantic values to 'objectively' and 'always'. However, this formal semantic theory does nothing at all to confer the needed understanding of the claimed lack of objectivity.

explanations of the A-theory/B-theory debate whose relation to T1 and T2 is not obvious.

(i) *Flow*

It is hard to make sense of the thought that *possible worlds* undergo “flow” or “passage”. Certainly the combination of propositional contingentism and modal elitism does not commit us to any such mysterious doctrine! This naturally suggests that the common A-theoretic slogan that *time* “flows” or “passes” has some content that goes beyond what is captured by the combination of propositional temporalism and temporal elitism.

Fine (2005c) thinks that this is a serious problem for that combination, or at least for its motivation. He expresses the concern as follows:

The standard realist [roughly: the propositional temporalist temporal elitist] faces a general difficulty. For suppose we ask: given a complete tenseless description of reality, then what does he need to add to the description to render it complete by his own lights? The answer is that he need add nothing beyond the fact that a given time *t* is present, since everything else of tense-theoretic interest will follow from this fact and the tenseless facts. But then how could this solitary ‘dynamic’ fact, in addition to the static facts that the antirealist is willing to accept, be sufficient to account for the passage of time? [The realist’s] conception of temporal reality. . . is as static or block-like as the anti-realist’s, the only difference lying in the fact that his block has a privileged center. Even if presentness is allowed to shed its light upon the world, there is nothing in his metaphysics to prevent that light being ‘frozen’ on a particular moment of time. (Fine 2005c, p. ***)³⁷

³⁷Note that Fine’s ‘anti-realists’ are not standard temporal B-theorists. While they are temporal egalitarians—as Fine puts it, they hold that ‘no time is privileged’—they accept propositional temporalism, which Fine regards as obviously true. Section 1.5 discussed Fine’s strategy for making sense of this combination.

I am not sure what Fine means by a ‘complete tenseless description of reality’. If ‘tenseless’ just means ‘necessarily either eternally true or eternally false’, the assumption that the proposition that according to the realist is a complete tenseless description of reality is one that ‘the anti-realist is willing to accept’ is highly problematic. The realist might very well think that temporal elitism itself is not only true, but necessarily and eternally true, while the anti-realist of course thinks that temporal elitism is false. Fine may intend ‘tenseless’ in some more demanding sense, but I don’t know what this sense might be, or why ‘everything else of tense-theoretic interest’ is supposed to follow from the facts that are tenseless in the *demanding* sense and the truth about which time is present.

To this, temporalist-elitists will reply that it is part of their view that the light of presentness is *not* ‘frozen on a particular moment of time’, since their view entails that other, future times *will* be present, and that other, past times *have been* present. What more could it conceivably mean for the light not be frozen? Fine anticipates this response, but finds it unconvincing on the following grounds:

However, the future presentness of $t+$ amounts to no more than t being present and $t+$ being later than t and, similarly, the past presentness of $t-$ amounts to no more than t being present and $t-$ being earlier than t . But then how can the passage of time be seen to rest on the fact that a given time is present and that various other times are either earlier or later than that time?

Given the standard way of thinking about instants (which section 1.3 attempted to lay out systematically), it is certainly true that the following two claims are necessarily equivalent:

- (9) a. $t+$ will be present
- b. $t+$ is after the instant that is present

(It is unclear whether this necessary equivalence justifies Fine’s ‘amounts to no more than’, which suggests that (9b) is in some sense “prior” to (9a); but let us not dwell on this.) Fine seems to find it mysterious how there *could* be any such necessary equivalence as this. At least, he seems to find it mysterious how (9a) and (9b) could be equivalent if ‘will be’ is given its ordinary meaning, the one that makes (9a) inconsistent with the idea that presentness is ‘frozen’ on an instant distinct from $t+$. Here are two instants, t and $t+$, which stand in a certain relation, *afterness*. How (the thought runs) could this relation, together with the fact that t is present, necessitate anything about where presentness will go next, if it goes anywhere? If presentness in fact is going to flow along the timeline at a rate of one second per second, how could that be a strictly necessary truth as opposed to a mere law of nature?

I’m not sure if I’m right to identify this as the thought behind Fine’s rhetorical question. But thoughts along these lines have, I think, been quite influential in the debate about the A-theory. Indeed, some A-theorists seem to have implicitly accepted that the equivalence between (9a) and (9b) is *not* metaphysically necessary, and engaged in strange and palpably wrong-headed metaphysical speculations (for example, speculations about the existence of multiple temporal dimensions) in trying to allow for the possibility

that things are otherwise. And the thought that A-theorists are forced to go down this route seems to be the basis of certain famous objections to the A-theory (e.g. Smart 1966). But it is hard to see where the mystery is supposed to lie. Why shouldn't (9a) and (9b) be metaphysically necessarily equivalent?

One way to make a putative metaphysical necessity seem unmysterious is to show how it follows logically from a 'real definition' or 'metaphysical analysis'—some claim of the form 'To be F is to be G', or 'For it to be the case that P is for it to be the case that Q'. In the case of (9a) and (9b), there are two well-known ways of doing this. We could, first, analyse 'after' in terms of tense operators, as follows:

- (10) For $t+$ to be after t is for it to be the case that always, if t is present, $t+$ *will* be present (**cite Prior**).

Alternatively, we could analyse tense operators in terms of 'at' and 'after':

- (11) For it to be the case that it will be that ϕ is for it to be the case that at some instant that is after the present instant, ϕ .

If we endorsed (10) or (11), there would be nothing mysterious about the necessary equivalence of (9a) and (9b). And what is supposed to be the principled objection to real definitions like (10) and (11)? Some putative real definitions are unbelievable on the grounds that they relate facts that are intuitively metaphysically distinct. But (10) and (11) are not at all like this. It is obvious that there is a very intimate relation between facts about how things are before and after the present and facts about how things will be and have been; I see nothing strange or repugnant about the suggestion that these two kinds of facts come to the same thing.

Thus, I can't see any intelligible further question about "flow" or "passage" that is not settled by the combination of propositional temporalism, temporal elitism, and the necessary equivalence of (9a) and (9b). Moreover, that necessary equivalence strikes me as something we have every reason to accept if we are going to go in for talk of instants at all.

(ii) *Objectivity*

TK

(iii) *Irreducibility*

Temporal elitism, as I have defined it, is a claim of *comparative* naturalness: the unit class of the present instant is much *more* natural than the unit classes of most other times and worlds. It is plausible that if this claim is true, the *property* of presentness must be a pretty natural property, more natural than almost any other property singling out an instant other than the present. But temporal elitists in my sense are not committed to the view that presentness is *perfectly* natural. Nor are they committed to the claim that presentness is in any other sense fundamental, or “irreducible”, or “metaphysically rock-bottom”, although they are of course free to endorse such further claims, as many philosophers in the A-theoretic tradition have done.³⁸ Explanations of the A-theory/B-theory debate sometimes make it sound as if some such further claim were constitutive of the A-theoretic perspective. But this is a mistake. For one thing, *accuracy*—the property of being an instant at which all and only the true propositions are true—would appear not to be fundamental or irreducible or rock-bottom, given that I have just defined it in terms of truth and the ‘at’ operator. And many distinguished philosophers who surely count as A-theorists have suggested that presentness just *is* accuracy.³⁹ More generally, many A-theorists have been opposed to the view that *instants of time* belong in our fundamental ontology, and have thus been eager to find a “reduction” of instants to something else, for roughly the same reasons that many philosophers have been sympathetic to a reductionist account of possible-worlds talk. One could for example reduce instants and worlds to sets of propositions, and propositions to set-theoretic constructions from objects and properties, and complex properties to set-theoretic constructions out of simpler properties, ultimately bottoming out in a handful of fundamental physical properties. This ambitious reductive project would go along with a correspondingly complex reductive account of properties like truth and accuracy, but is not foreign to the spirit of the A-theory.

Another “irreducibility” claim that is sometimes included in characterisations of the A-theory is the view that some *tense operators*—e.g. the operators ‘It has been the case that’ and ‘It will be the case that’—are fundamental (primitive, irreducible, perfectly natural, “structural” . . .). Such claims about the status of the tense operators have certainly been quite popular within

³⁸(*cite Deasy, and check Deasy for further references*)

³⁹*Cite: Prior, Fine, Markosian, Crisp . . .

the A-theoretic tradition since the foundational work of Prior (*cite*). They have been particularly attractive to those A-theorists who are suspicious of the ontology of instants of time, since the most obvious candidates to play the role of “reductions” are schemata like those in group (A) which involve quantification over instants (e.g. ‘It will be the case that ϕ iff at some future instant, ϕ ’). But others who seem to belong to the tradition have not shared in this suspicion, and have been happy to treat such schemata as reductions of the other tense operators to the operator ‘At t ’, which could itself be treated as primitive or further reduced. It strikes me as wrong-headed to think of this fine-grained dispute about metaphysical priority as marking the central fault line in the space of views about the metaphysics of time. The ideology of reduction and fundamentality is not all that clear, especially for operators; and disputes about exactly how the arrows of reduction point within some tightly-knit cluster of notions are rarely of great significance for the rest of philosophy. If we were determined to invoke the ideology of reduction or fundamentality in characterising the A-theory, we would do much better to go for something like ‘Some fundamental facts hold temporarily’, or ‘Some fundamental properties are instantiated temporarily’, or ‘Not all facts can be reduced to eternal facts’. A-theorists can entertain all sorts of interesting reductionist projects so long as they do not attempt to bridge the crucial gap between the temporary and the eternal.

(iv) *Anti-contextualism*

TK

1.7 Summary and preview

The main goal of this chapter was to introduce and elucidate two pairs of questions—T1 and T2, and M1 and M2. The elucidation consisted largely in a spelling out of certain background assumptions about propositions (section 1.2), instants, and worlds (section 1.3). In dealing with theorists who reject these assumptions, we must proceed warily: in some cases, we will want to say that certain people who give different answers to my questions are really in agreement about the deep underlying issues, their disagreement being merely a reflection of their idiosyncratic views about the ontology of propositions, instants, and worlds. Often, in engaging with such theorists, we will do best to look past what they say using the words ‘proposition’,

‘world’ and ‘instant’, and to ask instead whether there are any entities they believe in that do play the roles outlined in sections 1.2 and 1.3, and consider their views about these entities, whatever they might be called.

Sections 1.4, 1.5 and 1.5 considered the prospects of views that combine propositional temporalism with temporal egalitarianism, or propositional contingentism with modal egalitarianism. They did not amount to a *re-futation* of these combinations. But they did raise a major challenge for proponents of the combined view, namely that of motivating a conception of naturalness in which the transient [contingent] properties of things play no role in making certain classes more natural than others, despite the fact that there seem to major differences between such properties as regards how easy they are to refer to and how much they make for similarity and dissimilarity, and despite the fact that some such properties must be included in any universal supervenience base.

Chapter 2 will be devoted to discussing a catalogue of arguments from the literature bearing on T1, T2, M1 and M2. I will make no attempt at completeness: starting with MacTaggart, the literature on this area has been filled with ambitious but failed arguments, and it would be the task of a whole career to point out the shortcomings in each of them. The arguments I do consider are of very different kinds — some draw on common sense, some on linguistics, some on physics. Most seem to me to be unsuccessful, or at least far from conclusive. But in seeing how they can be resisted, we will learn a lot about the best ways of developing the views introduced in the present chapter, and uncover some interesting and hitherto unexplored corners of logical space.